

The upper limb exhibits an exceptional range of movement, which permits such wide positioning of the hand that it acts as a powered, mobile, sensory organ. This range is enabled by a number of factors. The thoracoscaphular and glenohumeral joints are remarkably little constrained and the muscles that both stabilize and move them are powerful. The skin is mobile, especially in the front of the neck, the axilla, across the joints, and on the dorsum of the hand. The deep fascial planes and the synovial tubes permit excursion of muscles and tendons against each other and adjacent structures, and the condensation of that fascia around the neurovascular bundles segregates them and enhances gliding. The excursion of the main nerves relative to the skeleton is increased by delicate vascular fascial adventitia and by gliding planes within the nerve trunks.

The innervation of the upper limb is complex and rich. There is a particular concentration of specialized sensory organelles in the wrist and hand, not only in the skin but also in the deep afferent system from muscles, tendons and joints. The postganglionic sympathetic vasomotor and sudomotor nerves are especially dense in the palmar skin. The basis of stereognosis, the ability to recognize the qualities of objects and textures, is a combination of stimuli from skin, tendon, muscle and joints. Movement is vital; blindfolded patients cannot identify the nature of an object if it is simply placed on the finger, but recognize it immediately if allowed to create spatial and temporal patterns by feeling the object or material between the moving finger and thumb.

BONES AND JOINTS

The bones of the upper limb include the scapula, clavicle, humerus, radius and ulna (connected by the interosseous membrane along most of their length), the eight bones forming the carpus, five metacarpals and fourteen phalanges (Figs 46.1–46.2). There are two phalanges in the thumb and three in each finger.

The thoracoscaphular joint is the platform for function of the upper limb; paralysis of either of the two great muscles, trapezius and serratus anterior, is crippling. The limb is directly attached to the axial skeleton by one bony articulation, the sternoclavicular joint. The glenohumeral joint is shallow and little constrained, whereas the acromioclavicular joint is a plane joint stabilized by strong ligaments. The range of movements provided by these joints and the muscles acting across them is enormous. Arthrodesis of the thoracoscaphular joint is effective in scapulohumeral dystrophy; it restores full elevation of the limb but protraction and retraction is lost. A correctly performed glenohumeral arthrodesis retains powerful adduction, protraction and medial rotation, but lateral rotation and most of elevation are lost. The elbow includes a hinge between the humerus and ulna, and two pivot joints: one between the humerus and radius, the other between the proximal radius and ulna. This permits a range of about 150° of extension and flexion, and 180° of pronosupination in conjunction with the distal radio-ulnar joint. The wrist complex allows a range of flexion and extension of about 140°, supplemented by some 70° of adduction and abduction. The condylar metacarpophalangeal joint enables 120° of flexion and extension, about 40° of abduction and adduction, and some rotation. Pronation and supination are unique to the primate upper limb. The range of movement of the thumb ray, which rests on the 'saddle' carpometacarpal joint, the qualities of the long muscles and the thenar muscles acting on the ray and the skin of the web space, are unique to humans.

SKIN AND FASCIA

The skin over the front of neck and posterior triangle is mobile and is sustained by underlying platysma. Large flaps may be raised safely when

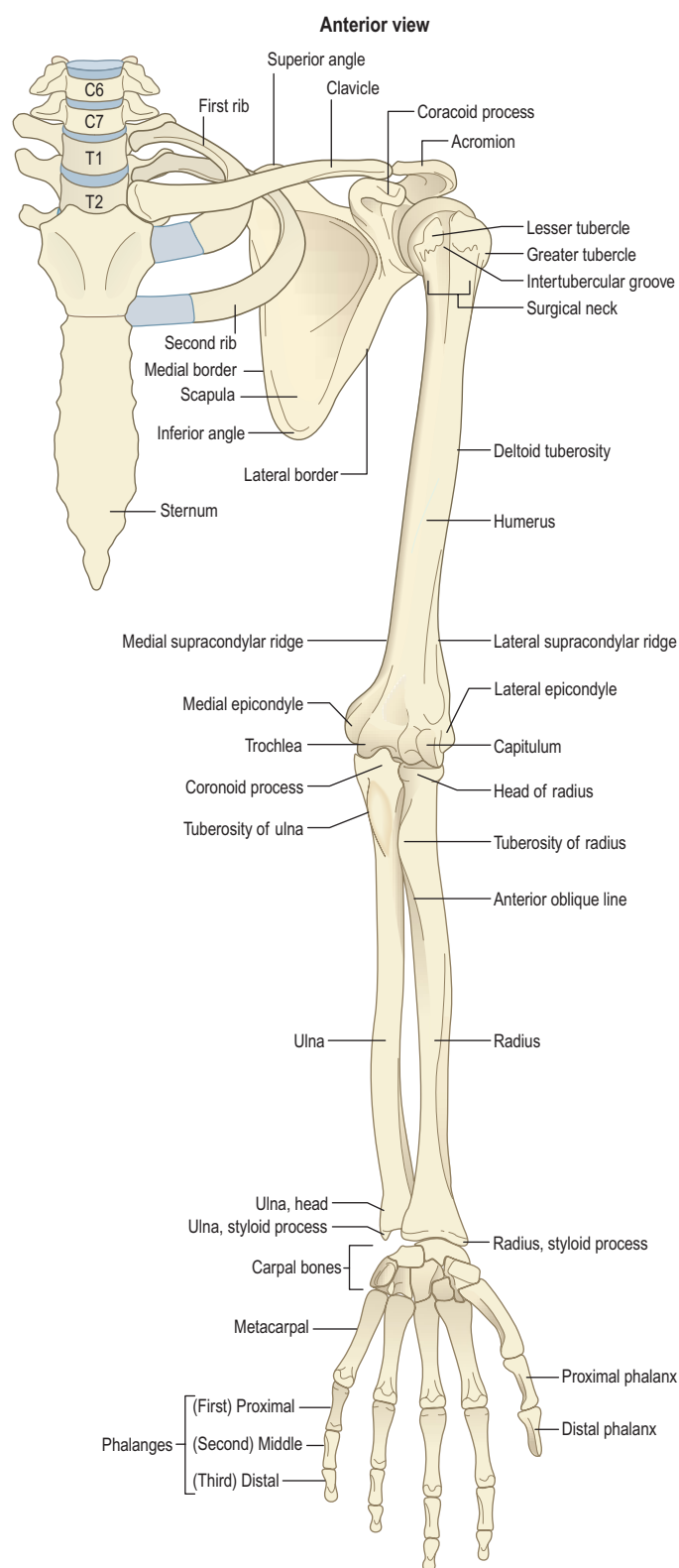


Fig. 46.1 The bones of the pectoral girdle and upper limb: anterior view.

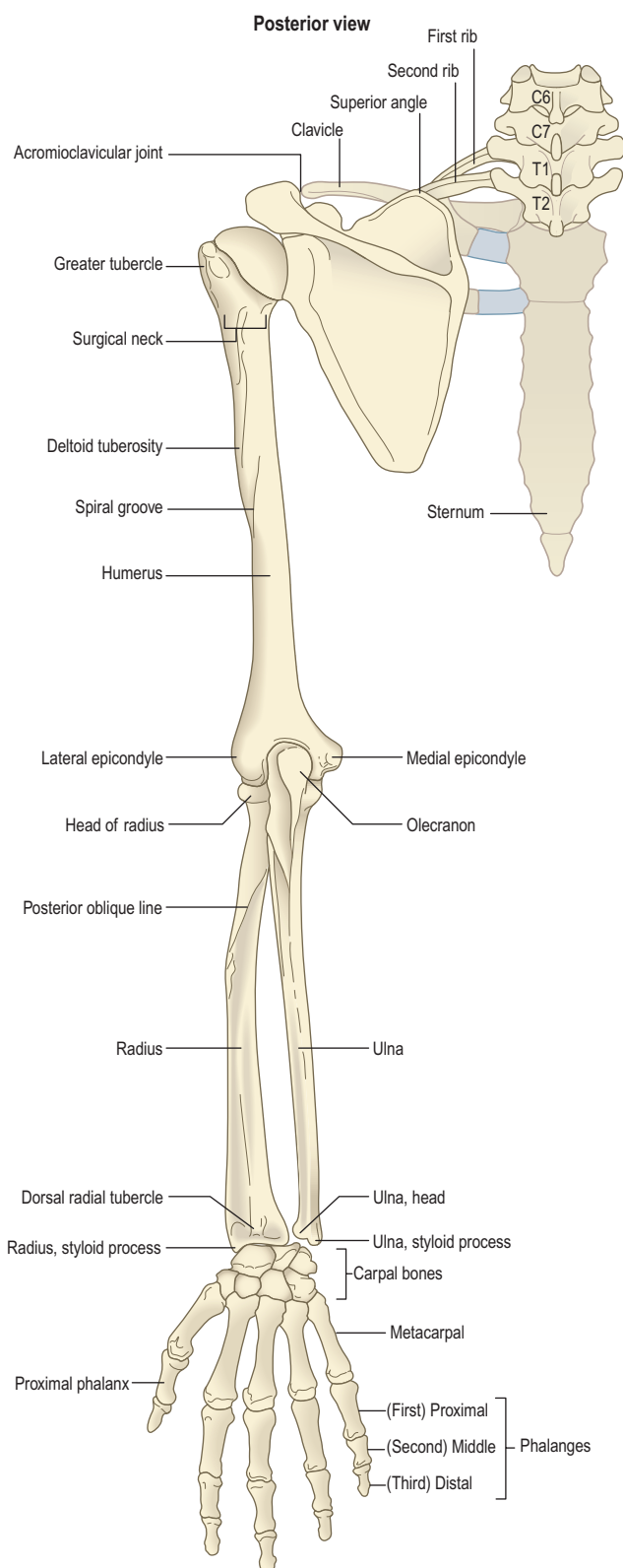


Fig. 46.2 The bones of the pectoral girdle and upper limb: posterior view.

platysma is included with the skin. The postaxial skin of the posterior aspect of the neck, shoulder, arm and forearm is thicker and hairy, whereas the glabrous preaxial skin on the anterior surface of the arm and forearm is thinner and more mobile. The situation is reversed in the hand, where the thick palmar skin is firmly secured by a fibrous skeleton to the palmar aponeurosis, whereas the dorsal skin is thinner and more mobile, especially across the joints. The characteristic furrows or creases at the elbow, wrist and the interphalangeal joints represent places of anchorage of the deep fascia. The hairy skin of the axilla is especially mobile, permitting the extensive range of movement at the glenohumeral joint; it is rich in sweat glands and their sympathetic nerves. The dense sympathetic innervation in the axilla, the hand and

forearm is important in temperature homeostasis. The consequences of scarring from burn or other injury, or badly placed incisions, are severe, and nowhere more so than after a deep burn of the axilla.

The superficial fascia (see [Chs 2 and 29](#)) is generally thicker on the dorsal aspect of the neck, shoulder, arm and forearm. Measurement of its thickness in the arm provides a useful measure of obesity. This fascia is an important gliding plane between the skin and the underlying deep fascia. Nerves and vessels are at risk of entrapment or even rupture where they perforate the deep fascia to ramify into the superficial fascia and skin.

The deep fascia, intermuscular septa and the interosseous membrane between the radius and ulna define discrete compartments that enable gliding of segregated structures against one another. The medial and lateral intermuscular septa of the arm and the interosseous membrane also provide wide areas for the attachment of muscles. The various fascial compartments are relevant to the spread of infection and tumour, and are especially important in ischaemia.

The prevertebral fascia (see [Fig. 29.4; p. 445](#)) envelops the phrenic nerve, scalene muscles, cervical primary rami, cervical sympathetic chain, and subclavian and vertebral arteries. It projects below the clavicle as the axillary sheath, enveloping the divisions and cords of the brachial plexus with the axillary artery, and continues into the arm as the brachial sheath, surrounding the brachial artery with its venae comitantes, the median nerve and the proximal part of radial and ulnar nerves. The great muscles of the shoulder and axilla – deltoid, pectoralis major and latissimus dorsi – are covered by deep fascia that winds around their deep surfaces to blend with the deep fascia of the arm. This latter is drawn upwards by suspensory condensations of fascia to form the cone-shaped axillary space. This arrangement is obvious to the reader who traces the anterior surface of pectoralis major with a finger, then follows it round and deep, before thrusting the finger upwards as far at least as the lateral part of the second rib.

The deep fascia of the arm and forearm forms a sort of sleeve, attached to the medial and lateral intermuscular septa in the arm, to the periosteum of the medial and lateral epicondyles and olecranon at the elbow, and to the periosteum of the ulna and radius ([Chs 48–49](#)). There are important condensations, such as the bicipital aponeurosis at the elbow, and the flexor and extensor retinacula at the wrist, which are, in turn, subdivided by septa. Discrete compartments with the forearm separate the superficial muscles from the deep. The reader will note the relative mobility of the extensor muscles that traverse the elbow: brachioradialis and the radial extensor muscles of the wrist form the ‘mobile wad’, overlying the deeper compartment containing the posterior interosseous nerve and vessels, supinator, the digital extensors, extensor carpi ulnaris and the long muscles acting on the thumb ray. Three compartments are found in the anterior aspect of the forearm. The deep flexor compartment contains the anterior interosseous nerve and vessels, flexor pollicis longus, flexor digitorum profundus and pronator quadratus. The superficial compartment contains the radial artery, pronator teres, flexor carpi radialis, palmaris longus and flexor digitorum superficialis. The ulnar nerve and vessels pass in a separate sheath close to the ulna. The anterior deep fascia of the forearm continues into the hand as the palmar aponeurosis, from which a complex arrangement of septa forms the fibrous skeleton of the hand (see [Fig. 50.7A](#)). The range of excursion of the main nerves of the upper limb across fixed points such as the first rib, the distal humerus and the distal radius is some 10–15 mm and is enabled by gliding between the adventitia and the epineurium. More movement occurs within the plane between the epineurium and perineurium, and also within the perineurium itself ([Commentary 9.1](#)).

Cutaneous innervation

Bundles of nerves enter the skin deep in the dermis and course towards the skin surface, giving off axons, nearly all unmyelinated, that innervate the associated end organs. The few myelinated axons terminate at hair follicles, Meissner corpuscles and Merkel complexes ([Ch. 7, Commentary 1.3](#)). The density of innervation of the epidermis is greatest in the proximal segment of the limb and there is little change between the twentieth and eightieth years.

MUSCLES

The sensorimotor cortex controls movements, not individual muscles. Reaching out to catch a flying object, such as a cricket ball, requires the coordination and integrated action of every muscle group in the upper limb, and indeed beyond. This system of complex and refined muscle

patterning depends on the integrity of the somatic afferent and efferent pathways in the central nervous system.

The muscles of the upper limb may be grouped according to their origin and the joints on which they act: (1) Muscles arising from the axial skeleton to act on the scapula include trapezius, levator scapulae, the rhomboids and serratus anterior. (2) Muscles arising from the axial skeleton to act on the glenohumeral joint include the sternal head of pectoralis major, pectoralis minor and latissimus dorsi. (3) Muscles passing between the scapula and the proximal humerus control the glenohumeral joint. They include supra- and infraspinatus, subscapularis, teres major and minor, and coracobrachialis. Deltoid and the clavicular head of pectoralis major also belong here, even though they arise in part from the clavicle. (4) The main muscles controlling the elbow include biceps and triceps; the long heads of both muscles traverse the glenohumeral joint to insert on the scapula. (5) The main muscles controlling supination and pronation are biceps brachii and supinator, pronator teres and pronator quadratus. (6) The radiocarpal joint is controlled by extensors carpi radialis longus and brevis, extensor carpi ulnaris, flexors carpi ulnaris and radialis, and palmaris longus. (7) The muscles acting on the thumb ray include the powerful flexor pollicis longus, the much weaker abductor pollicis longus, and extensors pollicis longus and brevis. These last three are essential for thumb function. The unopposed action of flexor pollicis longus leads to the virtually useless thumb in palm posture. (8) The extension and flexion of the metacarpophalangeal and interphalangeal joints of the fingers rest on coordinated activity in extensor digitorum, flexors digitorum superficialis and profundus, and the interosseous and lumbrical muscles. Extensor digitorum alone extends the metacarpophalangeal joints; the long flexors alone flex the interphalangeal joints. The imbalance caused by paralysis of the small muscles leads to 'clawing' deformity (see Fig. 46.27). (9) The small muscles of the hand may be considered as those controlling the thumb ray and web space; adductor pollicis lies deep to the first dorsal interosseous; abductor and flexor pollicis brevis and opponens pollicis are superficial and form the ball of the thumb. The interosseous muscles acting with the lumbricals flex, abduct and adduct the metacarpophalangeal joints and, in conjunction with the long flexor and extensor muscles, enable full extension of the proximal interphalangeal joints. Abductor and opponens digiti minimi and flexor digiti minimi brevis form the hypothenar eminence and act on the little finger.

Many muscles act on more than one joint; thus, the long heads of biceps and triceps flex and extend the glenohumeral joint, as well as the elbow. In addition, biceps brachii is a powerful supinator of the forearm. Extensor carpi radialis longus not only extends the wrist, but also is a powerful abductor of that joint and a flexor of the elbow. Flexor carpi ulnaris is the most powerful muscle in the forearm; it contributes to elbow flexion, flexes and adducts the wrist, and is active in all sustained movements of the wrist.

Certain muscles are functionally segregated, e.g. the anterior part of deltoid is a powerful flexor of the glenohumeral joint, whereas the posterior part is the most powerful extensor of that joint.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

The detailed regional descriptions of these systems are found in Chapters 48, 49 and 50.

ARTERIES

The axial vessel is the subclavian artery, which arises from the brachiocephalic trunk on the right and directly from the arch of the aorta on the left (Fig. 46.3). The artery is described in three parts that are successively anterior, deep and lateral to scalenus anterior. The second and third parts are in close relation to the primary ventral rami of C7, C8 and T1, and to the middle and lower trunks of the brachial plexus. The subclavian artery becomes the axillary artery at the posterior margin of the first rib. The axillary artery is closely related to the divisions of the brachial plexus deep to the clavicle, and to the cords below it. It continues deep to pectoralis minor and becomes the brachial artery at the inferior margin of teres major.

The brachial artery is closely related to the median nerve; both move from the medial side of the arm to the anterior aspect of the elbow, lying medial to the tendon of biceps brachii and deep to the bicipital aponeurosis. The artery divides into the radial and ulnar arteries just distal to the elbow. The ulnar artery is consistently the larger of the two. The common interosseous artery arises close to its origin and subsequently divides into the anterior and posterior interosseous arteries. The

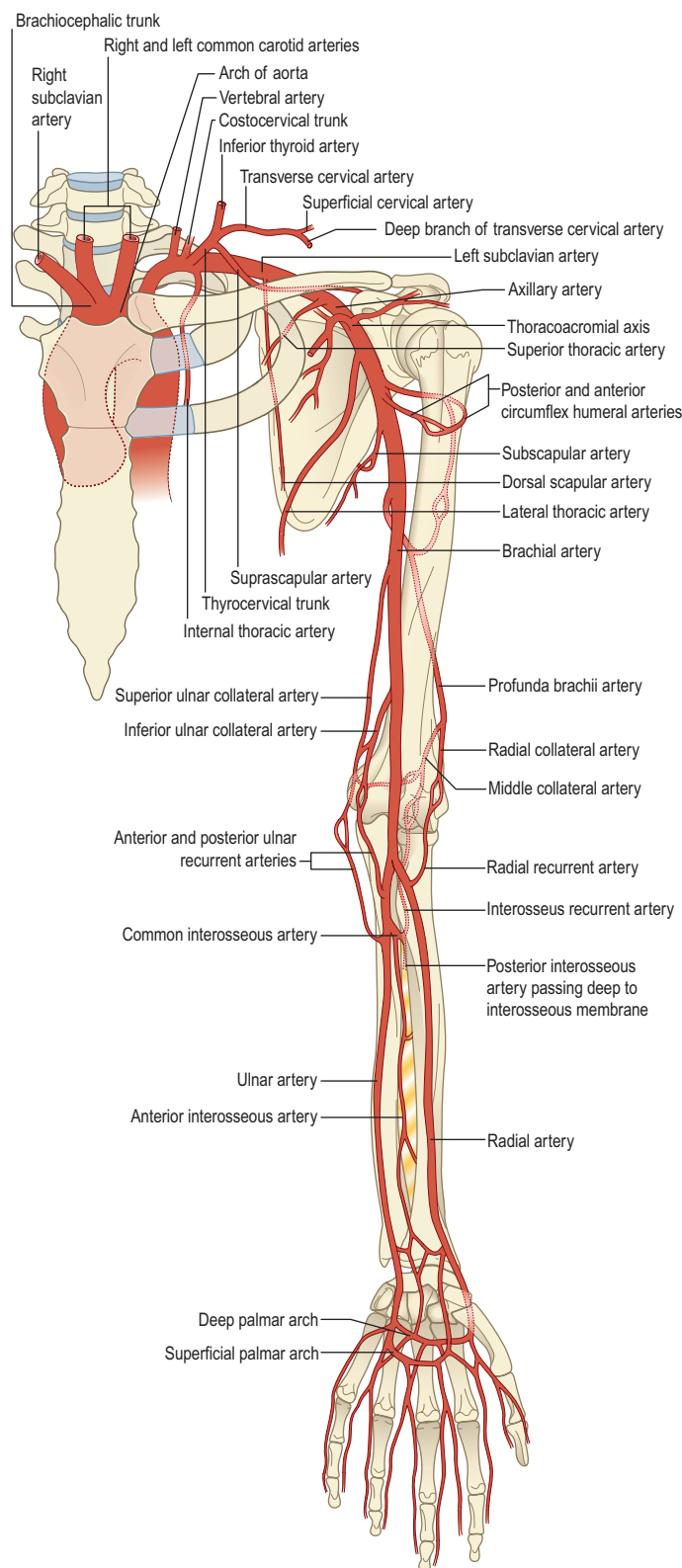


Fig. 46.3 The arteries of the upper limb.

anterior interosseous artery, accompanied by the anterior interosseous nerve, lies on the interosseous membrane in the deepest part of the deep flexor compartment. The posterior interosseous artery is separated from the membrane by the deep extensor muscles. The radial and ulnar arteries remain in the flexor compartment of the forearm, the ulnar artery moving towards the ulnar nerve in the proximal quarter of the forearm; it is usually the dominant vessel for the hand. At the wrist, the radial artery passes dorsally and crosses the scaphoid and trapezium.

Important branches from the main axial vessels form extensive anastomoses that provide a collateral circulation. For example, the collateral circulation formed by branches of the thyrocervical trunk with the circumflex humeral and subscapular arteries permits survival of the limb

after occlusion of either the third part of the subclavian artery, or of the axillary artery deep to the clavicle and pectoralis minor. The profunda brachii is an important channel for the posterior muscles of the arm. It accompanies the radial nerve and contributes to the collateral circulation about the elbow with the ulnar collateral and recurrent vessels and the radial collateral and recurrent vessels. The anterior interosseous artery is effectively an end artery. A series of anastomoses between the radial and ulnar arteries, such as the transversely orientated palmar and dorsal arches and the superficial and deep palmar arches, maintain a rich blood supply to the wrist and hand. There are extensive interconnections between these arches and between the dorsal and palmar phalangeal arteries.

Pulses

The pulsation of the subclavian artery is palpable at the lateral margin of sternocleidomastoid and is easily blocked by digital pressure against the first rib. This simple manoeuvre has saved lives and limbs, and should be widely known and rehearsed because it is the best method for emergency control of bleeding from the deeply placed axillary artery. The method was used successfully for high amputation or disarticulation of the shoulder in the Napoleonic wars.

The brachial artery is palpable in the arm in the groove, or valley, between biceps brachii anteriorly and the medial head of triceps brachii posteriorly; it can be traced down to the anterior aspect of the elbow, where it moves towards the midline. The radial and ulnar pulses are palpable at the wrist, where the vessels emerge from under the cover of overlying muscles.

VEINS

The upper limb is drained by superficial and deep groups of vessels.

The superficial group starts as an irregular dorsal arch on the back of the hand. The cephalic vein begins at the radial extremity of the arch, and ascends along the lateral aspect of the arm within the superficial fascia to enter the deltopectoral groove. It pierces the deep fascia above the superior margin of pectoralis minor and enters the axillary vein near the tip of the coracoid (Fig. 46.4). The basilic vein drains the ulnar end of the arch, passes along the medial aspect of the forearm, pierces the deep fascia at the elbow, and joins the venae comitantes of the brachial artery to form the axillary vein. The prominent median cubital vein links the cephalic and basilic veins on the flexor aspect of the elbow. It receives a number of tributaries from the flexor aspect of the forearm and gives off the deep median vein, which pierces the fascial roof of the antecubital fossa to join the venae comitantes of the brachial artery.

The deep group of veins drains the tissues beneath the deep fascia of the upper limb and is connected to the superficial system by perforating veins. The deep veins accompany the arteries, usually as venae comitantes, ultimately becoming the axillary, and subsequently the subclavian, vein.

The subclavian vein is the central continuation of the axillary vein. It starts at the inferior margin of the first rib, which it crosses. Initially deep to the clavicular head of pectoralis major and then to subclavius, which is a buffer between it and the overlying medial portion of the clavicle, the vein then runs deep to the insertion of sternocleidomastoid. Sometimes, only the superior margins of the vein rise above the medial end of the clavicle; sometimes, the entire vein does so. The phrenic nerve and the inferior part of scalenus anterior lie posteriorly. The subclavian vein joins the internal jugular vein to form the brachiocephalic vein at the medial border of scalenus anterior, behind the sternoclavicular joint (see Fig. 29.14). A pair of valves usually lies within 2 cm of this junction. The subclavian vein receives, as tributaries, the external jugular, dorsal scapular and, sometimes, the anterior jugular veins. The trifurcation of subclavian, external and internal jugular veins is demonstrable by a forced expiration (Valsalva) manoeuvre, when the distended veins rise up above the sternoclavicular joint, filling the fossa above the jugular notch of the sternum (Video 46.1). Main lymphatic vessels join the subclavian veins at, or very close to, this junction with the internal jugular vein. The left subclavian vein receives the thoracic duct; the right receives the right lymphatic duct (see Fig. 29.16). There is a risk of air embolism after wounding of the subclavian, external or internal jugular veins.

Cutaneous blood supply

The blood supply of the skin of the upper limb is generally abundant (Fig. 46.5) and rather more robust than that in the lower limb.

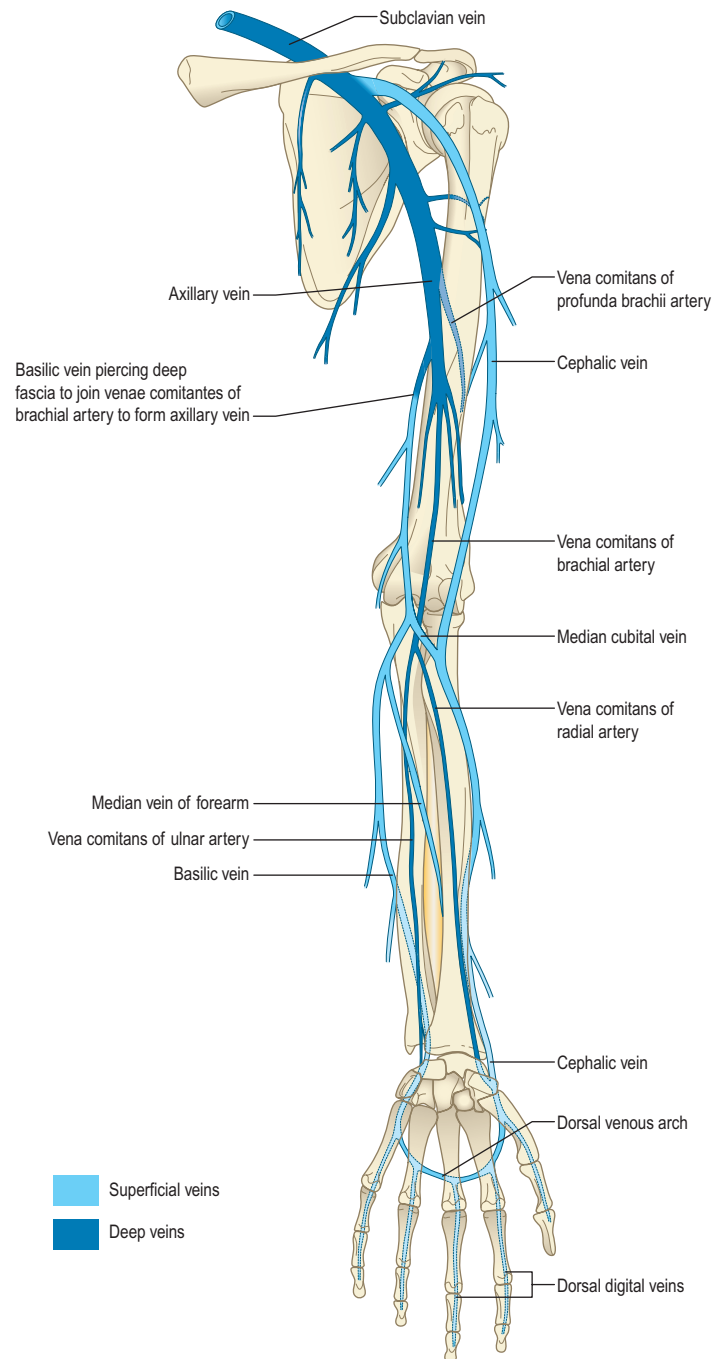


Fig. 46.4 The veins of the upper limb.

Proximally based flaps of skin caused by 'degloving' injury are more likely to survive; even a distally based flap of palmar skin may do so. The skin of the lower leg is notoriously sensitive to the effects of fracture because of rupture of the vessels perforating the deep fascia; this problem is less severe in the upper limb. Three interconnected pathways may be recognized in the upper limb, namely: muscle perforators, deep fasciocutaneous vessels and the direct cutaneous supply.

Muscle perforators



Available with the *Gray's Anatomy* e-book

Fasciocutaneous system



Available with the *Gray's Anatomy* e-book

Direct cutaneous supply



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Vessels pass from deep axial vessels through muscle, perforate the deep fascia and supply the overlying skin. The latissimus dorsi myocutaneous flap based on the thoracodorsal vessels was an important development in plastic and reconstructive surgery, not only for the replacement of lost skin or reconstruction of the breast, but also in the evolution of the free-functioning muscle transfer.

The fasciocutaneous system rests on branches that arise from deep vessels and pass along intermuscular septa to perforate the deep fascia and ramify over its surface; extensive areas of skin are supplied in this way. The development of fasciocutaneous flaps, either as pedicle flaps or free grafts, has been a revolutionary advance in the treatment of severe wounds; not only is it possible to replace large areas of lost skin, but also the improvement of the tissue bed that overlies the main nerves enhances regeneration and relieves neuropathic pain.

The direct cutaneous supply is exemplified in the palmar skin, where branches from the underlying digital vessels pass directly into the overlying skin.

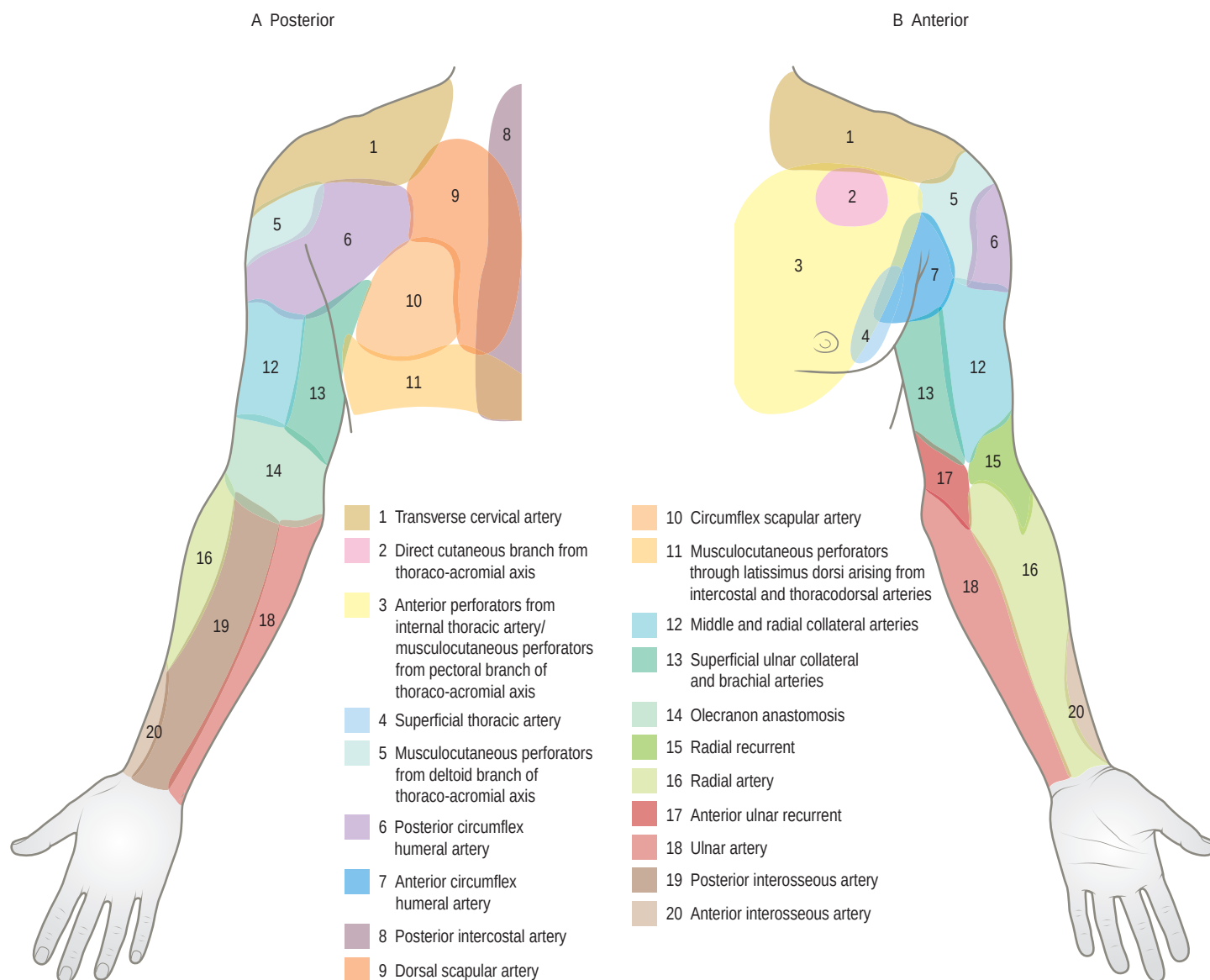


Fig. 46.5 The anatomical territories served by the cutaneous blood supply to the upper limb. (With permission from Cormack GC, Lamberty BGH 1994 *The Arterial Anatomy of Skin Flaps*, 2nd ed. Edinburgh: Churchill Livingstone.)

LYMPHATIC DRAINAGE

Superficial tissues

Superficial lymphatic vessels begin in cutaneous plexuses. In the hand, the palmar plexus is denser than the dorsal plexus. Digital plexuses drain along the digital borders to their webs, where they join the distal palmar vessels, which pass back to the dorsal aspect of the hand. The proximal palm drains towards the carpus, medially by vessels that run along its ulnar border, and laterally to join vessels draining the thumb. Several vessels from the central palmar plexus form a trunk that winds round the second metacarpal bone to join the dorsal vessels that drain the index finger and thumb.

In the forearm and arm, superficial vessels run with the superficial veins. Collecting vessels from the hand pass into the forearm on all carpal aspects. Dorsal vessels, after running proximally in parallel, curve successively round the borders of the limb to join the ventral vessels. Anterior carpal vessels run through the forearm parallel with the median vein of the forearm to the cubital region, then follow the medial border of biceps brachii before piercing the deep fascia at the anterior axillary fold to end in the lateral axillary lymph nodes.

Lymph vessels that lie laterally in the forearm receive vessels that curve round the lateral border from the dorsal aspect of the limb. They follow the cephalic vein to the level of the tendon of deltoid, where most incline medially to reach the lateral axillary nodes; a few continue with the vein and drain into the infraclavicular nodes. Vessels lying medially in the forearm are joined by vessels that curve round the medial border of the limb and follow the basilic vein. Proximal to the elbow, some end in supratrochlear lymph nodes whose efferents, together with the medial vessels that have bypassed them, pierce the

deep fascia with the basilic vein and end in the lateral axillary nodes or deep lymphatic vessels.

Collecting vessels from the deltoid region pass round the anterior and posterior axillary folds to end in the axillary nodes. The scapular skin drains either to subscapular axillary nodes or by channels that follow the transverse cervical vessels to the inferior deep cervical nodes.

Deeper tissues

Deep lymph vessels follow the main neurovascular bundles (radial, ulnar, interosseous and brachial) to the lateral axillary nodes. They are less numerous than the superficial vessels and communicate with them at intervals. A few lymph nodes occur along the vessels. Scapular muscles drain mainly to the subscapular axillary nodes, and pectoral muscles drain mainly to the pectoral, central and apical nodes.

Efferent vessels from the deep cervical nodes form the jugular trunk. The right jugular trunk drains the right upper limb, the right half of the thorax and the right head and neck, and enters the right thoracic duct or directly enters the right subclavian vein close to its junction with the internal jugular. The left jugular trunk usually enters the thoracic duct, but it may directly enter either the subclavian or internal jugular vein (see Figs 29.16, 48.45).

INNERVATION

The accessory nerve, dorsal scapular nerve, long thoracic nerve (nerve to serratus anterior), suprascapular nerve and nerve to subclavius all innervate muscles that act on the scapulothoracic 'joint' and therefore will be summarized here.

Accessory nerve

The integrity of the accessory nerve is fundamental to thoracoscaphic function and essential for scapulohumeral rhythm (Camp and Birch 2011). The intraspinal and intracranial course of the accessory nerve and of the segment of the nerve that lies in the anterior triangle are described on page 467.

The accessory nerve passes either deep to or through sternocleidomastoid, to enter the posterior triangle in a consistent and important relation to the ascending nerves of the cervical plexus. The great auricular nerve is the key to exposure of the accessory nerve, which emerges, usually as one trunk, 5–10 mm cephalad to the point where the great auricular nerve winds around sternocleidomastoid, and moves from a plane posterior to the muscle to one that is anterior. The general relations of the posterior triangle vary with individual physique, but this relation between the two nerves is reliably consistent. The nerve now runs across the fatty areolar tissue at the apex of the posterior triangle, in close relation to the superior superficial cervical lymph nodes deep to platysma. It is separated from the underlying levator scapulae by this areolar tissue, and then, more deeply, by the prevertebral fascia. One slender branch passes to the upper fibres of trapezius either deep to sternocleidomastoid or just beyond it. The nerve pierces the fascia covering the deep surface of trapezius close to the anterior border of the muscle and then runs caudally about 2 cm from that border. The nerve passes down in a characteristic sinuous fashion, accompanied by slender vessels to the deep, inner face of trapezius. A branch from the cervical plexus (C3, 4) joins the nerve just above the clavicle. Intraoperative stimulation of this branch rarely evokes a muscular response; it probably conveys afferent fibres from the middle and lower parts of trapezius. The nerve now turns medially, continuing in parallel to the spine of the scapula about 4 cm above it. At the medial end of the spine, the nerve turns caudally again, passing down parallel to the medial border of the scapula. It divides into terminal muscular branches about 7 cm distal to the scapular spine. Although sternocleidomastoid, and the middle and the lower portions of trapezius, may be partially supplied by branches from the cervical plexus, the upper part of trapezius is innervated solely by the accessory nerve.

Dorsal scapular nerve

The dorsal scapular nerve arises above the clavicle from the proximal segment of the ventral primary ramus of C5. It passes posteriorly, piercing scalenus medius, to run down in the plane between levator scapulae and serratus posterior superior and the posterior scalene muscles. It continues along the anterior border of the rhomboid muscles about 1.5 cm medial to the vertebral border of the scapula and is closely related to the dorsal scapular artery. The dorsal scapular nerve innervates the rhomboid muscles and, together with branches from C3 and C4, it supplies levator scapulae.

Long thoracic nerve (nerve to serratus anterior)

The long thoracic nerve is formed by branches that arise above the clavicle from the proximal segments of the ventral primary rami of C5, C6 and C7. It innervates serratus anterior, a muscle that is essential for the function of the thoracoscaphic joint. The muscular branches from the rami join deep to scalenus medius, and the trunk passes down posterolateral to the muscle, on the floor of the posterior triangle deep to the suprascapular nerve. The branches from C5 and C6 are the largest. The nerve follows a sinuous course deep to the investing fascia covering the anterior faces of the digitations of serratus anterior. It is accompanied here by a branch of the thoracodorsal artery, and trends posterolaterally towards the mid-axillary line.

Suprascapular nerve

The suprascapular nerve (C5, 6) usually arises as the first branch of the upper trunk but it frequently springs directly from the ventral primary ramus of C5. It innervates supra- and infraspinatus.

Nerve to subclavius

The slender nerve to subclavius (C5, 6) springs from the upper trunk and passes anteriorly. It descends anterior to the plexus and the subclavian artery and passes above the subclavian vein to supply subclavius.

Brachial plexus

The muscles, joints and skin of the upper limb are innervated by the ventral primary rami of C5, C6, C7 and C8, and nearly all of the ventral primary ramus from T1, collectively forming the brachial plexus (Figs 46.6–46.7). A branch from C4 to C5, which contributes to the innervation of the glenohumeral and elbow flexor muscles, is seen occasionally

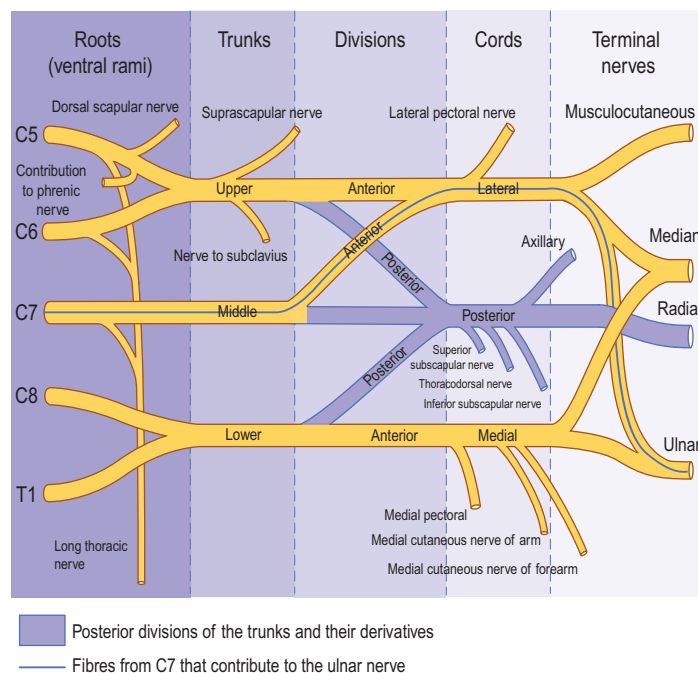


Fig. 46.6 A schematic plan of the brachial plexus. (Adapted with permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

at operation. A muscular contribution from T2 is rare. The rami enter the posterior triangle of the neck between scalenus anterior and medius. Rami from C7 and C8 are the largest (C8 contains about 30,000 myelinated axons) and those from C5 and T1 are the smallest (between 15,000 and 20,000 myelinated axons). An adult brachial plexus contains between 120,000 and 150,000 myelinated axons, of which 25% innervate the shoulder girdle and glenohumeral joint. The proportion of motor fibres is greatest in C5 and C8; the sensory contribution is greatest in C7. A complex interchange of branches, before the main nerves of the upper limb are formed, produces the trunks, divisions and cords of the brachial plexus.

Blood supply The blood supply of the brachial plexus is derived from vessels arising from the subclavian and vertebral arteries. Important branches pass from the vertebral artery to the rami of C5 and C6 and the more proximal cervical nerves. Branches from the costocervical trunk provide a rich supply to T1 and C8. Extensive contributions come from the suprascapular and superficial cervical arteries that arise from the thyrocervical trunk. In at least one-third of cases, the superficial cervical and dorsal scapular arteries arise from the thyrocervical trunk as the transverse cervical artery. In clinical terms, the arteries arising from the thyrocervical trunk have become the lifeline to the upper limb and must be preserved during operations in cases when the ruptured subclavian artery has not been repaired. The dorsal scapular artery may arise from the third part of the subclavian artery to pass between the upper and middle trunks of the brachial plexus.

Trunks of the brachial plexus

The upper trunk is formed by C5 and C6, where these nerves emerge from deep to scalenus anterior. The middle trunk is the continuation of C7. The lower trunk is formed by C8 and T1, where these nerves cross anterior to the first rib. The ramus of T1 takes an upward course across the deep face of the neck of the first rib behind the pleura and the vertebral and subclavian arteries towards the lower trunk. The formation of the trunks is fairly consistent; they lie in front of one another rather than side by side, with the subclavian artery passing anteromedially. The phrenic nerve crosses C5 to pass anteromedially on the surface of scalenus anterior. The upper trunk, its divisions and the suprascapular nerve can all be palpated in the supraclavicular fossa in a subject of normal physique. The examining finger identifies first the subclavian pulse and then the nerves, whilst rolling the fingertip laterally (see Video 46.1).

Divisions of the brachial plexus

The trunks divide into anterior and posterior divisions. The upper trunk divides 2–3 cm above the clavicle, and the divisions of the middle and lower trunk are formed deep to the clavicle. The posterior divisions of the upper and middle trunks are consistently larger than their anterior

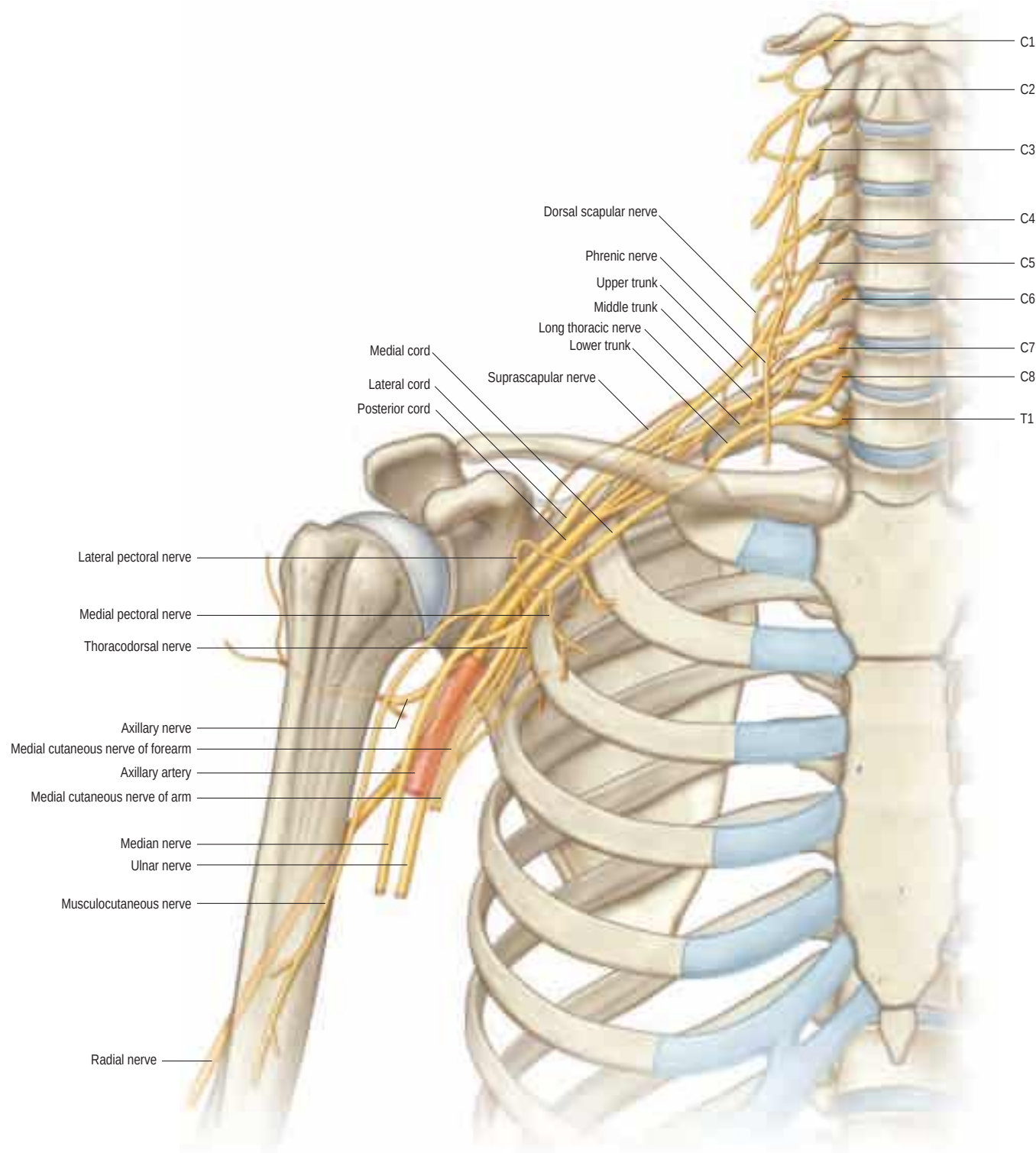


Fig. 46.7 The brachial plexus. Note the sequence: ventral (anterior) primary rami; trunks; divisions; cords; nerves. The trunks are upper, middle and lower, and the cords are lateral, medial and posterior, according to their position in relation to the axillary artery, which is variable. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

divisions. The posterior division of the lower trunk is consistently smaller than the anterior division and is absent in about 10% of cases.

Cords of the brachial plexus

The cords are formed by the confluence of divisions: the lateral cord from the anterior divisions of the upper and middle trunks; the posterior cord by all three posterior divisions; and the medial cord by the anterior division of the lower trunk and, sometimes, by a branch from the anterior division of the middle trunks. The divisions of the trunks and the formation of the cords represent an important anatomical and functional differentiation. The posterior divisions and posterior cord innervate postaxial (extensor) musculature; the anterior divisions and the lateral and medial cords innervate preaxial (flexor) musculature.

The formation and relations of the three cords are variable and, indeed, their designations somewhat misleading. Immediately inferior to the clavicle, the posterior cord is lateral, the medial cord is posterior, and the lateral cord is anterior, in relation to the axillary artery; the cords assume their appropriate relations about the axillary artery deep to pectoralis minor. There is considerable variation in this arrangement; most commonly, the axillary artery lies anterior to the three cords and the median nerve.

The branches of the posterior cord, the largest of the three trunks, are consistent. In sequence, they are the upper subscapular, thoracodorsal, lower subscapular, axillary and radial nerves. The branches of the medial cord are usually predictable; the medial pectoral nerve and medial cutaneous nerve of the forearm are succeeded by the division into the medial root of the median nerve and the ulnar nerve. The ulnar

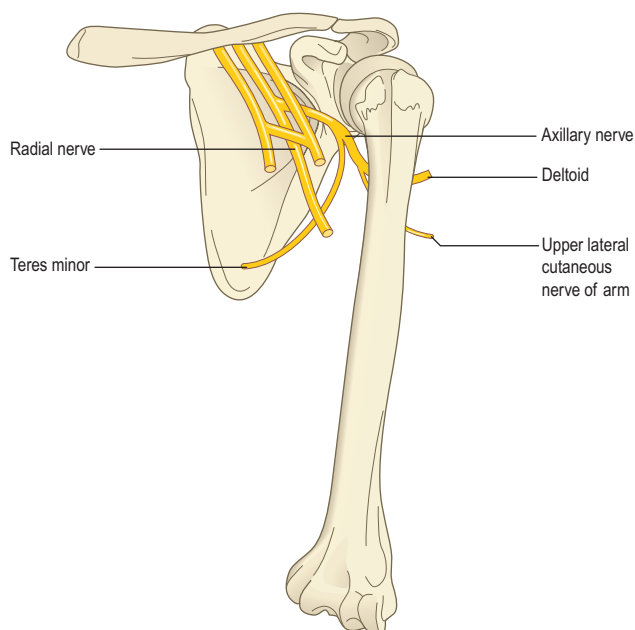


Fig. 46.8 Motor and sensory branches of the axillary nerve. (With permission from O'Brien M, *Aids to the Examination of the Peripheral Nervous System*, 5th ed, 2010, Edinburgh: Saunders, Elsevier.)

nerve may arise as two or three branches. The greatest variation in formation of trunk nerves is found within the lateral cord. Occasionally, the musculocutaneous nerve arises more distally than usual, springing either directly from the lateral cord as two or three branches or even from the median nerve itself. Sometimes, the highest of these branches enters coracobrachialis no more than 2 or 3 cm below the coracoid process. The lateral root of the median nerve may arise as two or three branches, and in some cases, it appears as a branch of the musculocutaneous nerve.

Axillary (circumflex) nerve

The axillary nerve (C5, 6) (Fig. 46.8) runs freely in loose fatty tissue in the axilla before turning around subscapularis to pass almost horizontally through the quadrilateral tunnel, where it is accompanied by the posterior circumflex humeral vessels (see Fig. 48.33). It innervates teres minor and deltoid, and the skin overlying deltoid on the lateral aspect of the shoulder.

Radial nerve

The radial nerve (C5, 6, 7, 8, T1) is the terminal branch of the posterior cord (Fig. 46.9). It is the largest nerve in the upper limb and the most commonly damaged. Between its origin at the level of the base of the coracoid and its entrance into the spiral groove, where it is accompanied by the profunda brachii artery, the radial nerve is supplied by fewer arteries than elsewhere along its course. The first 8–10 cm of the nerve may not have a nutrient artery and will, therefore, be relatively avascular if the nerve is transected at its origin. The nerve lies closest to the bone where it pierces the lateral intermuscular septum to pass through a short tunnel bounded by bone and unyielding fascia. The nerve is tethered here, a common level of rupture, entrapment or compression.

The nerve to the medial head of triceps arises in the axilla and accompanies the main nerve as it passes through the intermuscular septum. The nerve(s) to the lateral head of triceps arise(s) within the spiral groove; that to the lateral head just before the radial nerve re-enters the anterior compartment. Two cutaneous nerves, the lower lateral cutaneous nerve of the arm and the posterior cutaneous nerve of the forearm, pass away from the main nerve in the final part of the spiral groove and course along the anterior face of the lateral head of triceps to perforate the deep fascia about four fingers' breadths above the lateral epicondyle. They innervate the skin of the lower lateral arm and the posterior aspect of the forearm.

The nerve to brachioradialis is formed about three fingers' breadths proximal to the lateral epicondyle; the nerve to extensor carpi radialis longus about one finger's breadth more distal; and that to extensor carpi radialis brevis about one finger's breadth above the epicondyle. These branches spring from the main nerve as it runs in the valley between brachialis anteriorly and brachioradialis posteriorly, which is a site where the nerve may be palpated. At the level of the tip of the lateral epicondyle, the radial nerve divides into the posterior interos-

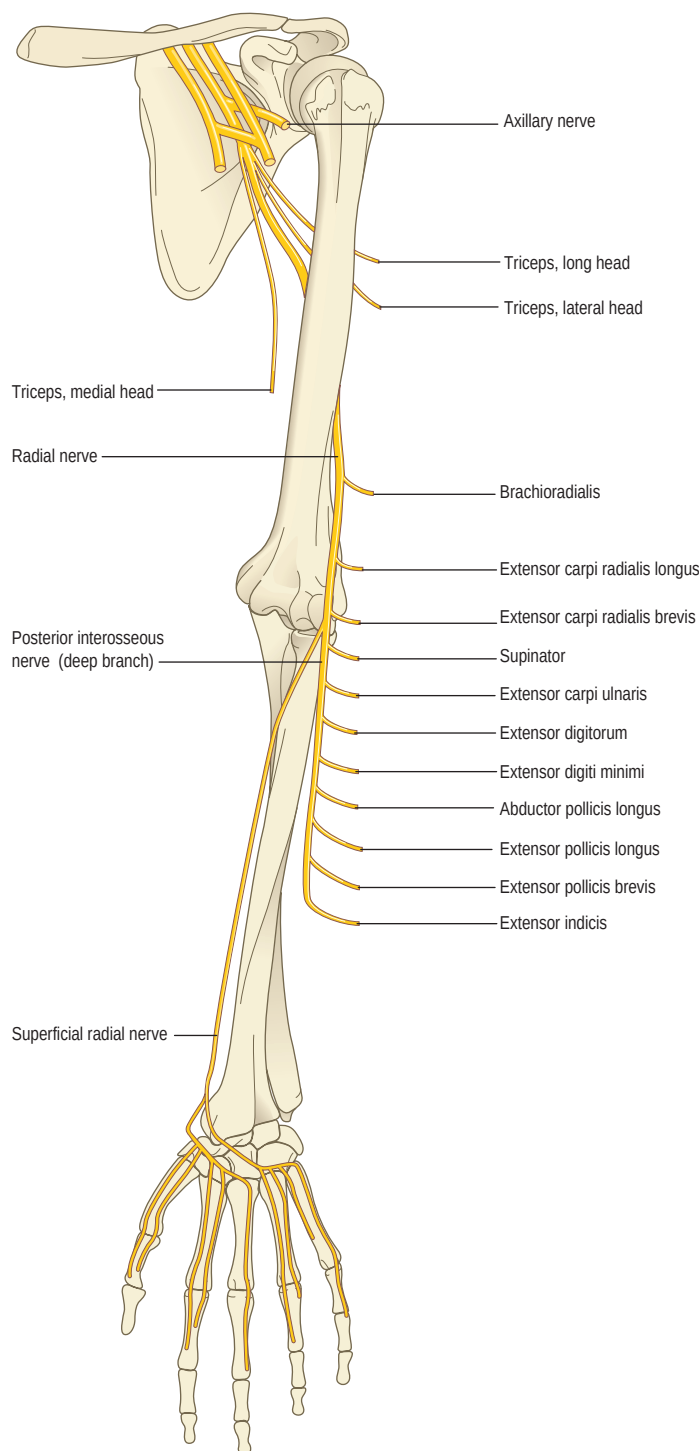


Fig. 46.9 Motor and sensory branches of the radial nerve. Variation exists in the cutaneous innervation of the dorsal aspects of the digits. Here, the radial nerve is shown to supply all five digits; the skin of the dorsum of the ring and little fingers is frequently innervated by the dorsal branch of the ulnar nerve. (With permission from O'Brien M, *Aids to the Examination of the Peripheral Nervous System*, 5th ed, 2010, Edinburgh: Saunders, Elsevier.)

seous and superficial radial nerves. Another branch to extensor carpi radialis brevis is usually given off here. The superficial radial nerve innervates the skin over the dorsum of the radial half of the ring, middle and index fingers, and that of the thumb as far as the distal interphalangeal joint. It often provides important sensation on the skin of the thumb web space and the skin on the adjacent sides of the thumb and index.

Median nerve

The median nerve is formed by the union of the lateral root (C6, 7) from the lateral cord, and the medial root (C8, T1) from the medial cord, which meet anterior to the third part of the axillary artery (Fig. 46.10). Fibres in the lateral root innervate the palmar skin of the thumb, index and most of the middle fingers, and pronator teres, flexor carpi

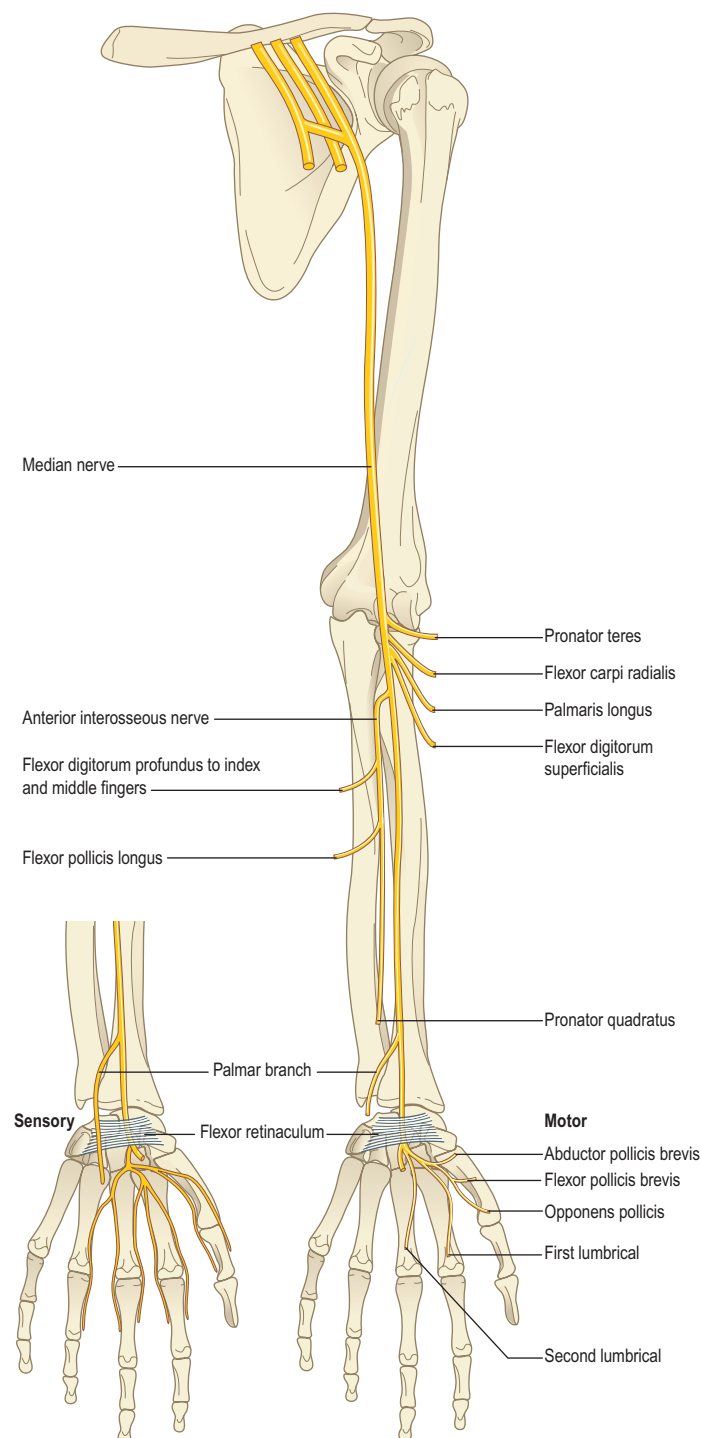


Fig. 46.10 Motor and sensory branches of the median nerve. Flexor pollicis brevis may be supplied by both median and ulnar nerves. (With permission from O'Brien M, *Aids to the Examination of the Peripheral Nervous System*, 5th ed, 2010, Edinburgh: Saunders, Elsevier.)

radialis and some of flexor digitorum superficialis (Fig. 46.11). The lateral root conveys most of the sympathetic fibres to the median distribution in the hand. The medial root carries fibres to the skin of the medial side of the middle and the lateral side of the ring finger, and also fibres to palmaris longus, flexor digitorum superficialis and the lateral part of flexor digitorum profundus, flexor pollicis longus, pronator quadratus and the median innervated muscles within the hand. The first branch of the median nerve, to pronator teres, arises 2–3 cm above the medial epicondyle. A second branch, or leash of branches, arises at the level of the tip of the medial epicondyle and innervates palmaris longus, flexor carpi radialis and flexor digitorum superficialis. Just proximal to the tendinous arcade of flexor digitorum superficialis, the median nerve gives off the anterior interosseous nerve. Accompanied by the anterior interosseous vessels, this large branch dives deeply to pass down along the interosseous membrane and supplies flexor pollicis longus, the radial half of flexor digitorum profundus and pronator

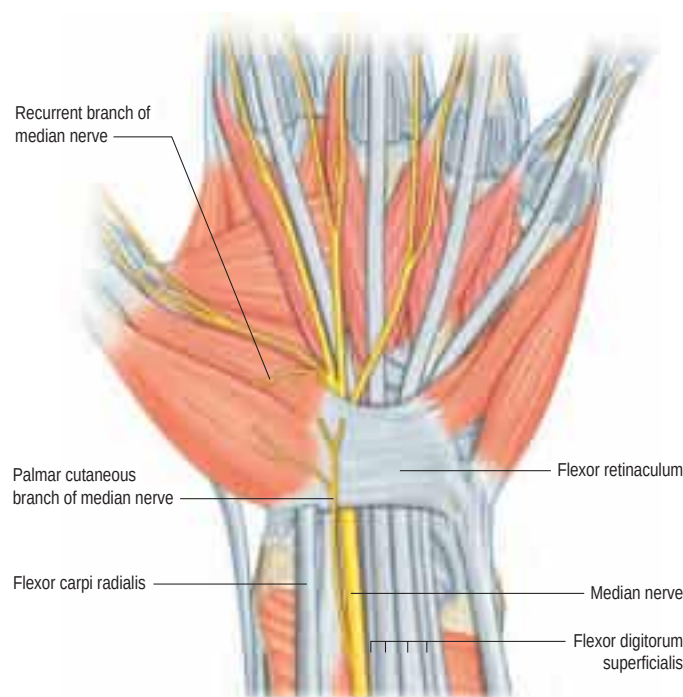


Fig. 46.11 The normal course of the median nerve and the palmar cutaneous nerve at the wrist. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

quadratus. There is no cutaneous distribution. The main nerve continues in the plane between the superficial and deep digital flexors, supplying two or three branches to the former. The palmar cutaneous nerve arises about 3 cm proximal to the proximal wrist crease, and passes lateral to the main nerve and superficial to the flexor retinaculum to innervate the skin of the proximal palm. The median nerve passes deep to the flexor retinaculum, into the carpal tunnel, to enter the palm. The nerve to the thenar muscles arises within, or just distal to, the tunnel, usually on the lateral side of the main nerve. The palmar digital nerves are formed within the palm of the hand.

The nerve is palpable along its course in the arm, where it is covered by deep fascia and skin, after emerging from deep to coracobrachialis. It is palpable on the flexor aspect of the elbow, deep to the bicipital aponeurosis, and is accompanied by the brachial artery in this segment of the limb. The median nerve is palpable at the wrist where it emerges from behind the superficial flexor tendons, just lateral to palmaris longus.

Ulnar nerve

The ulnar nerve (C7, C8, T1) is the continuation of the medial cord (Fig. 46.12). It inclines posteromedially to perforate the medial intermuscular septum about 10 cm proximal to the medial epicondyle and passes through a fibrous canal, the arcade of Struthers, which is between 5 and 6 cm long. The walls of the canal include the medial intermuscular septum and the fascial sheath investing the medial head of triceps brachii. The nerve next passes behind the medial epicondyle, then between the capsule of the humero-ulnar joint and the overlying arcuate ligament, which joins the two origins of flexor carpi ulnaris, i.e. the nerve passes through the cubital tunnel. It is accompanied by the superior collateral ulnar vessels in the lower third of the arm as far as the entrance to the cubital tunnel, and then by the posteroinferior ulnar collateral vessels. The anastomosis between these vessels permits the use of most of the ulnar nerve as a free vascularized graft, in cases where the parent nerves, C7, C8 and T1, are irreparable. In the forearm, the nerve runs with the ulnar vessels in a well-defined sheath deep to the anterior margin of flexor carpi ulnaris. It divides into superficial (sensory) and deep (motor) components at the wrist, both passing through Guyon's canal.

The superficial division innervates the skin of the medial two digits and palmaris brevis. The deep branch passes, with the deep branch of the ulnar artery, between abductor and flexor digiti minimi, pierces opponens digiti minimi and then runs across the deep palmar space with the deep palmar arch. It innervates all the muscles of the hand, except for the first and second (index and middle) lumbricals, abductor pollicis brevis, opponens pollicis and part of flexor pollicis brevis.

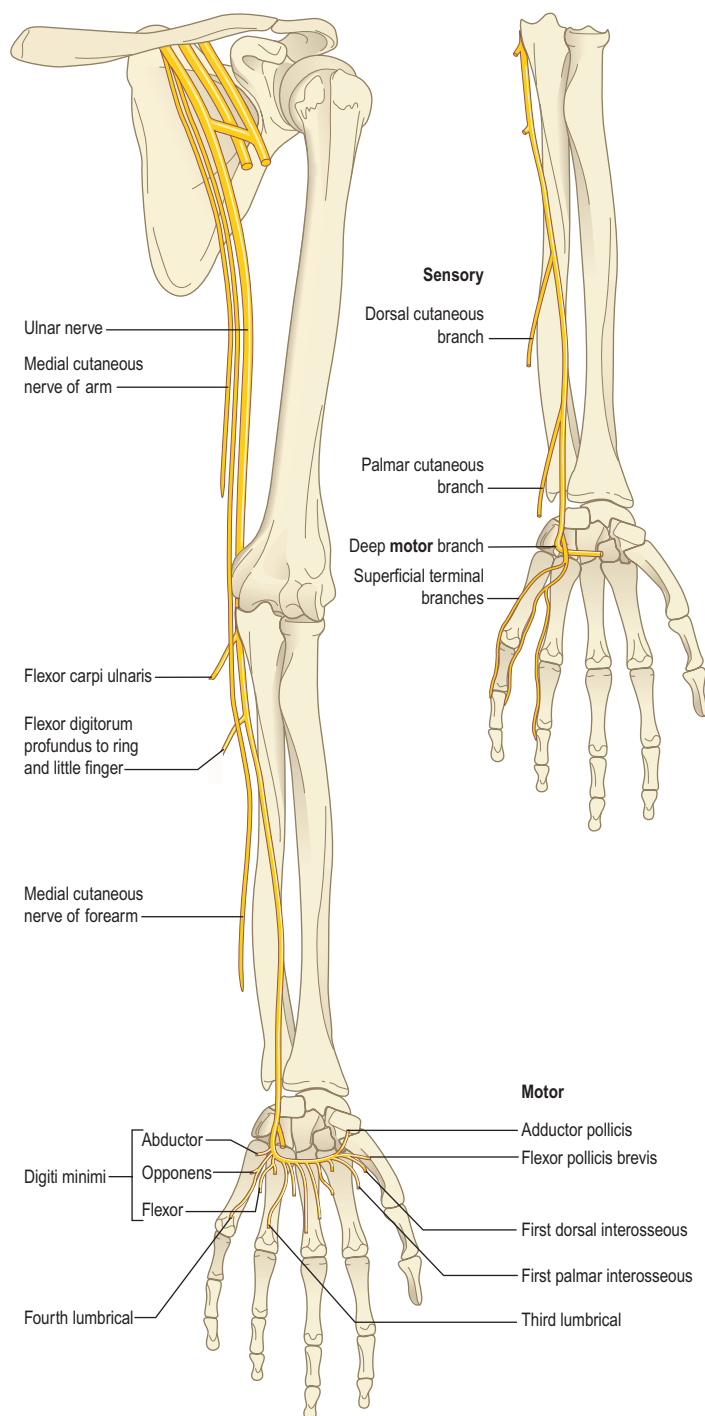


Fig. 46.12 Motor and sensory branches of the ulnar nerve and the medial cutaneous nerves of the arm and the forearm. Flexor pollicis brevis may be supplied by both median and ulnar nerves. (With permission from O'Brien M, *Aids to the Examination of the Peripheral Nervous System*, 5th ed, 2010, Edinburgh: Saunders, Elsevier.)

A branch to the medial head of triceps is often given off in the distal third of the arm. There are usually two neurovascular pedicles to flexor carpi ulnaris, the first at the exit from the cubital tunnel, the second about one-quarter of the way down the forearm. Several branches pass to flexor digitorum profundus. The dorsal cutaneous nerve arises about 6 cm proximal to the ulnar styloid, winds around the ulna, at first deep to flexor carpi ulnaris, and then subcutaneously. It innervates the skin of the dorsum of the hand and little finger, and the medial side of the ring finger.

The ulnar nerve is palpable in the lower arm and behind the medial epicondyle, where it is subcutaneous. There is an important relation here: a posterior branch of the medial cutaneous nerve of the forearm sweeps across the cubital tunnel and gets in the way of medial surgical exposures. Inadvertent injury may cause severe pain. The ulnar nerve is palpable at the wrist where it emerges from under flexor carpi ulnaris.

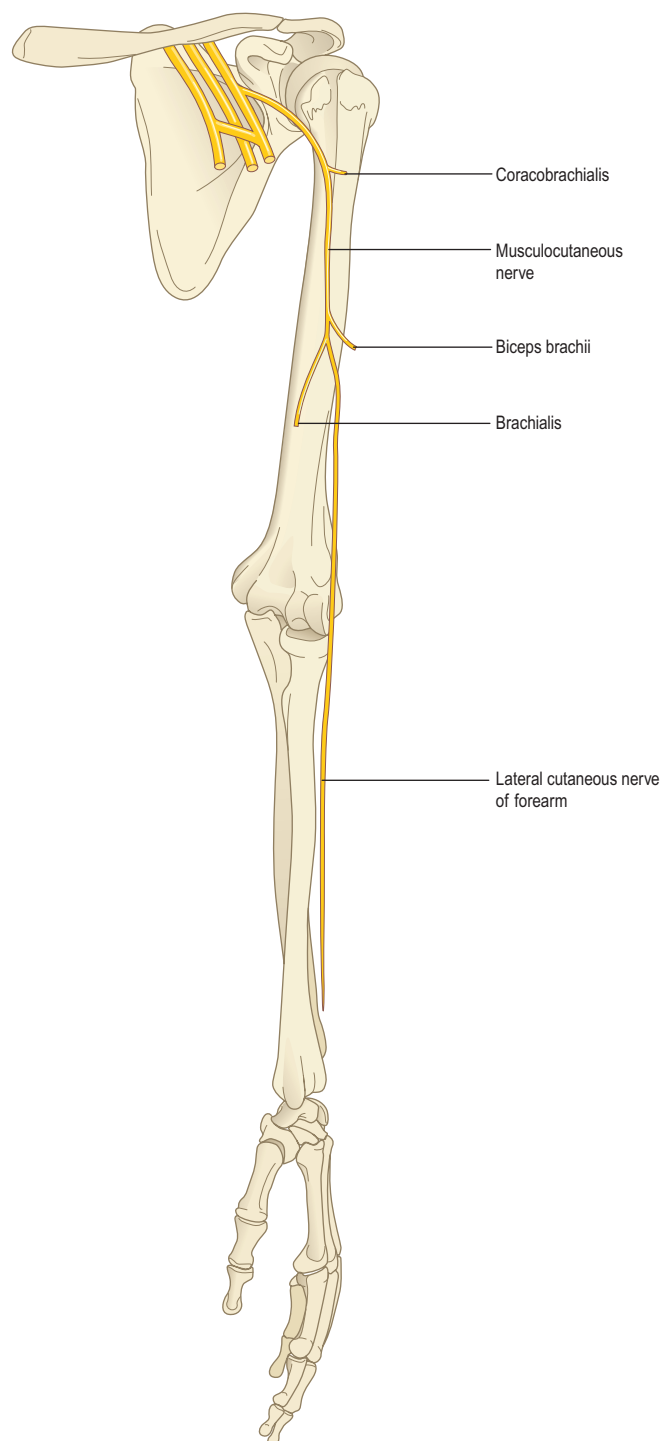


Fig. 46.13 Motor and sensory branches of the musculocutaneous nerve. (With permission from O'Brien M, *Aids to the Examination of the Peripheral Nervous System*, 5th ed, 2010, Edinburgh: Saunders, Elsevier.)

The dorsal cutaneous branch can be palpated in those of slender physique where it crosses the subcutaneous ulna.

Musculocutaneous nerve

The musculocutaneous nerve (C5, 6, 7) is one of the two terminal branches of the lateral cord, the other being the lateral root of the median nerve (Fig. 46.13). This division usually occurs at the level of the third part of axillary artery, where the lateral cord is lateral to the vessel, but it may occur more proximally, where the cord is anterior to the vessel. The nerve may also arise as two or three branches from the cord, or even from the median nerve. It supplies coracobrachialis, both heads of biceps brachii and most of brachialis, and continues as the lateral cutaneous nerve of forearm, deep to biceps, before emerging lateral to it about 11 cm above the lateral epicondyle, and then running down the lateral aspect of the forearm. The nerve sometimes innervates the skin over the first metacarpal and the lateral part of the thenar eminence.

Segmental innervation of the upper limb

There is a segmental pattern to the innervation of the upper limb. The most proximal muscles are supplied by branches of the uppermost rami, and the most distal muscles are supplied by branches derived from C8 and T1. The segmental pattern of innervation is shown more clearly in the cutaneous supply. The cervical supply has been, as it were, extruded from the supply to the trunk. Thus, in the transition of cutaneous innervation from the skin of the neck to that of the trunk, there is a change from the fourth cervical to the second thoracic segment anteriorly, and a change from the fifth cervical to the first thoracic segment posteriorly (Figs 46.14–46.15; see Fig. 16.10 for other variations). (For further reading, see Ladak et al (2014).) The truly autonomous area of cutaneous supply of each main component nerve is small and variable in extent and location.

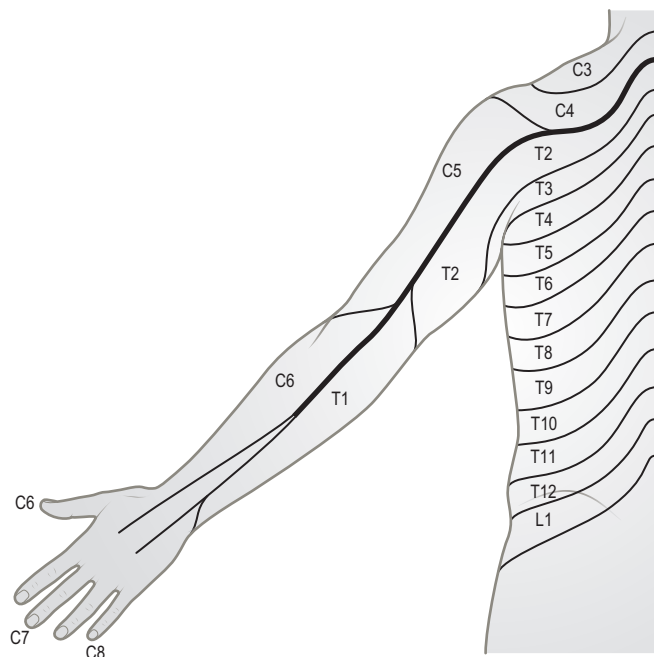


Fig. 46.14 Approximate distribution of the dermatomes on the posterior aspect of the upper limb. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

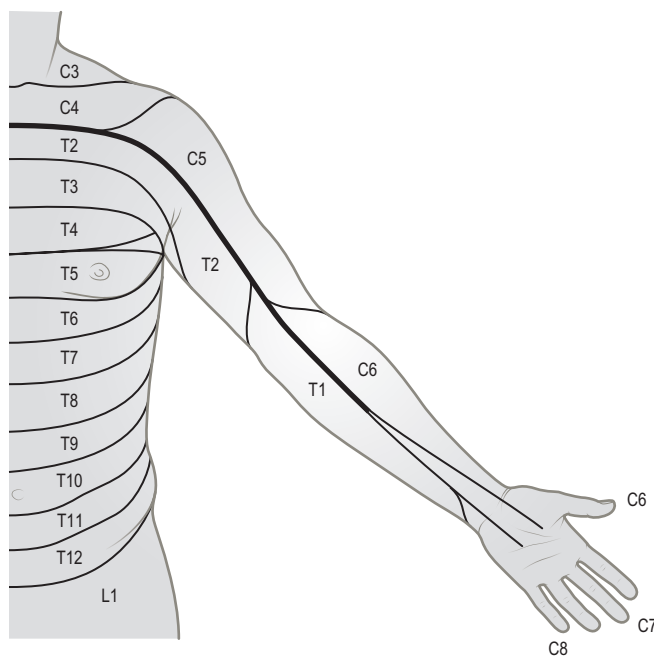


Fig. 46.15 Approximate distribution of the dermatomes on the anterior aspect of the upper limb. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

C5 regularly controls extension, abduction and lateral rotation of the shoulder. C6 usually innervates biceps brachii, brachialis, lateral head of triceps, brachioradialis, extensor carpi radialis longus, and sometimes contributes to the innervation of pronator teres, supinator and flexor carpi radialis. C7 gives widespread innervation throughout the limb; in those unusual instances where it alone is damaged, the patient will note a rather diffuse loss of function throughout the upper limb, often without complete anaesthesia and sometimes without paralysis of any single significant muscle group. C7 consistently supplies latissimus dorsi. C8 innervates the extensors of the digits and of the thumb in at least one-third of cases. T1 innervates these muscles and also the medial head of triceps in at least 10%. Muscles, movements and segmental innervation in the upper limb are set out in Tables 46.1–46.3. Table 46.4 helps to distinguish between upper and lower motor neurone lesions.

Autonomic nerves

The preganglionic sympathetic inflow to the upper limb is derived from neurones in the intermediolateral horn of spinal cord segments T2–6/7. Myelinated preganglionic axons pass from the spinal nerves to the thoracic sympathetic chain via white rami communicantes. They pass up the chain to the middle cervical and inferior cervical (stellate or cervico-thoracic) ganglia, where they relay.

The unmyelinated postganglionic fibres are distributed by grey rami communicantes to the nerves of origin of the brachial plexus. C5 and C6 receive grey rami from the middle cervical ganglion; C7 and C8 receive grey rami from the inferior cervical ganglion; T1 not only receives a grey ramus from the inferior cervical ganglion but also contributes a white ramus to it (see Figs 16.13, Fig. 46.16). Interruption of this white ramus is one cause of Horner's (Bernard–Horner) syndrome: drooping eyelid, narrow palpebral fissure, a contracted pupil, and vasomotor with sudomotor paralysis in the ipsilateral face and neck. The sympathetic outflow to the upper limb passes with the peripheral nerves and is particularly rich in the palmar skin of the hand and in the skin of the axilla.

CLINICAL DIAGNOSIS OF FOCAL NERVE LESIONS IN THE UPPER LIMB



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ACUTE ISCHAEMIA AND COMPARTMENT SYNDROMES IN THE UPPER LIMB



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SURFACE ANATOMY

SKELETAL LANDMARKS

The clavicle is both visible and palpable from its prominent and relatively rounded medial (sternal) end, which forms the lateral boundary of the suprasternal notch, to its flattened lateral end, where it joins the acromion (Fig. 46.29A). The acromioclavicular joint line is palpable as an anteroposteriorly aligned ridge or groove. The anterior, superior and lateral surfaces of the acromion are subcutaneous and palpable (Fig. 46.29C). Posteriorly, the palpable acromial angle marks the junction of the acromion with the crest of the scapular spine. From this point, the spine of the scapula, which is subcutaneous and easily visible in thin subjects, is palpable to the medial (vertebral) border of the scapula, where it lies opposite the spine of the third thoracic vertebra.

The medial border of the scapula, superior to the level of the scapular spine, is covered by trapezius and, therefore, difficult to palpate. Below the spine, the medial border is palpable down to the inferior angle. Although covered by teres major and latissimus dorsi, the inferior angle is palpable when approached inferiorly, and can be seen and/or palpated as it moves around the thoracic wall during shoulder movements. The inferior angle is opposite the spine of the seventh thoracic vertebra and overlies the seventh rib; it is a convenient landmark from which the ribs can be counted along the lateral chest wall, e.g. when a thoracotomy is being performed.

Table 46.1 Movements, muscles and segmental innervation in the upper limb

Joint	Movement	Muscle	Innervation	C3	C4	C5	C6	C7	C8	T1
SCAPULA	ELEVATION	Upper trapezius	Accessory n.							
		Levator scapulae	Dorsal scapular n.							
	DEPRESSION	Lower trapezius	Accessory n.							
	RETRACTION	Middle trapezius	Accessory n.							
SHOULDER	PROTRACTION	Serratus anterior	Long thoracic n.							
		Anterior deltoid	Axillary n.							
	FLEXION	Pectoralis major (clavicular head)	Medial and lateral pectoral nn.							
		Pectoralis major (sternocostal head)	Medial and lateral pectoral nn.							
		Coracobrachialis	Musculocutaneous n.							
		Posterior deltoid	Posterior deltoid							
	EXTENSION	Infraspinatus	Suprascapular n.							
		Teres minor	Axillary n.							
		Teres major	Lower subscapular n.							
		Latissimus dorsi	Thoracodorsal n.							
	VERTICAL ABDUCTION	Middle deltoid	Axillary n.							
	VERTICAL ADDUCTION	Supraspinatus	Suprascapular n.							
		Pectoralis major (sternocostal head)	Medial and lateral pectoral nn.							
		Latissimus dorsi	Thoracodorsal n.							
		Coracobrachialis	Musculocutaneous n.							
	HORIZONTAL ABDUCTION	Posterior deltoid	Axillary n.							
	HORIZONTAL ADDUCTION	Pectoralis major (clavicular head)	Medial and lateral pectoral nn.							
		Pectoralis minor	Medial and lateral pectoral nn.							
		Anterior deltoid	Axillary n.							
	MEDIAL ROTATION	Subscapularis:	Upper and lower subscapular nn.							
		Teres major	Brachial plexus							
		Latissimus dorsi	Thoracodorsal n.							
		Anterior deltoid	Axillary n.							
	LATERAL ROTATION	Infraspinatus	Suprascapular n.							
		Teres minor	Axillary n.							
		Posterior deltoid	Axillary n.							
ELBOW	FLEXION	Biceps brachii	Musculocutaneous n.							
		Brachialis	Musculocutaneous and radial nn.							
		Brachioradialis	Radial n.							
	EXTENSION	Triceps brachii	Radial n.							
	SUPINATION*	Biceps brachii	Musculocutaneous n.							
		Supinator	Posterior interosseous n.							
	PRONATION*	Pronator quadratus	Anterior interosseous n.							
WRIST	FLEXION	Flexor carpi radialis	Median n.							
		Palmaris longus	Median n.							
		Flexor carpi ulnaris	Ulnar n.							
	EXTENSION	Extensor carpi radialis longus	Radial n.							
		Extensor carpi radialis brevis	Posterior interosseous n.							
		Extensor carpi ulnaris	Posterior interosseous n.							
	ABDUCTION	Extensor carpi radialis longus	Radial n.							
		Extensor carpi radialis brevis	Posterior interosseous n.							
		Flexor carpi radialis	Median n.							
	ADDUCTION	Extensor carpi ulnaris	Posterior interosseous n.							
		Flexor carpi ulnaris	Ulnar n.							
FINGERS	FLEXION (MP/PIP joints)	Flexor digitorum superficialis	Median n.							
		Dorsal interossei	Ulnar n.							
		Palmar interossei	Ulnar n.							
	FLEXION (DIP joints)	Flexor digitorum profundus (lateral)	Anterior interosseous n.							
		Flexor digitorum profundus (medial)	Ulnar n.							
	FLEXION (MP joint)	Flexor digiti minimi brevis	Ulnar n.							
	EXTENSION (MP/PIP/DIP joints)	Extensor digitorum	Posterior interosseous n.							
		Extensor indicis	Posterior interosseous n.							
	EXTENSION (MP/PIP/DIP joints)	Flexor digiti minimi	Posterior interosseous n.							
	EXTENSION (PIP/DIP joints)	Lumbricals I and II	usually Median n.							
		Lumbricals III and IV	usually Ulnar n.							
	ABDUCTION	Dorsal interossei	Ulnar n.							
	ABDUCTION (thumb fixed)	Abductor pollicis brevis	Median n.							
	ABDUCTION	Abductor digiti minimi	Ulnar n.							
THUMB	ADDUCTION	Palmar interossei	Ulnar n.							
	OPPOSITION	Opponens digiti minimi	Ulnar n.							
	FLEXION (IP joint)	Flexor pollicis longus	Anterior interosseous n.							
	FLEXION/ROTATION (MP joint)	Flexor pollicis brevis	Median n. and/or ulnar n.							
	EXTENSION (MP joint)	Extensor pollicis brevis	Posterior interosseous n.							
	EXTENSION (IP joint)	Extensor pollicis longus	Posterior interosseous n.							
	ABDUCTION	Abductor pollicis longus	Posterior interosseous n.							
	ABDUCTION/ROTATION	Abductor pollicis brevis	Median n.							
	ADDUCTION/ROTATION	Adductor pollicis	Ulnar n.							
	ADDUCTION/FLEXION (MP joint)	Palmar interossei I	Ulnar n.							
	OPPOSITION	Opponens pollicis	Median n. and ulnar n.							

Muscles and movements that have diagnostic value are marked in red. Spinal roots have been given the same colour (light peach, dark peach, red) when they innervate a muscle to a similar extent. Red has been used to indicate roots from which there is known to be a dominant contribution.

*These movements occur at the radio-ulnar joints.

Abbreviations: DIP, distal interphalangeal; IP, interphalangeal; MP, metacarpophalangeal; PIP, proximal interphalangeal.

Table 46.2 Segmental innervation of the muscles of the upper limb

Nerves	Muscles
C3, 4	Trapezius, levator scapulae
C5	Rhomboids, deltoids, supraspinatus, infraspinatus, teres minor, biceps
C6	Serratus anterior, latissimus dorsi, subscapularis, teres major, pectoralis major (clavicular head), biceps, coracobrachialis, brachialis, brachioradialis, supinator, extensor carpi radialis longus
C7	Serratus anterior, latissimus dorsi, pectoralis major (sternal head), pectoralis minor, triceps, pronator teres, flexor carpi radialis, flexor digitorum superficialis, extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi
C8	Pectoralis major (sternal head), pectoralis minor, triceps, flexor digitorum superficialis, flexor digitorum profundus, flexor pollicis longus, pronator quadratus, flexor carpi ulnaris, extensor carpi ulnaris, abductor pollicis longus, extensor pollicis longus, extensor pollicis brevis, extensor indicis, abductor pollicis brevis, flexor pollicis brevis, opponens pollicis
T1	Flexor digitorum profundus, intrinsic muscles of the hand (except abductor pollicis brevis, flexor pollicis brevis, opponens pollicis)

Table 46.3 Segmental innervation of joint movements of the upper limb

Territory	Muscles	Nerves
Shoulder	Abductors and lateral rotators Adductors and medial rotators	C5 C6–8
Elbow	Flexors Extensors	C5, 6 C7, 8
Forearm	Supinators Pronators	C6 C7, 8
Wrist	Flexors and extensors	C6, 7
Digits	Long flexors and extensors	C7, 8
Hand	Intrinsic muscles	C8, T1

Table 46.4 Movements and muscles tested to determine the location of a lesion in the upper limb

Arm movement	Muscle	Upper motor neurone	Root	Reflex	Nerve
Shoulder abduction	Deltoid	++	C5		Axillary
Elbow flexion	Biceps Brachioradialis		C5/6 C6	+	Musculocutaneous Radial
Elbow extension	Triceps	+	C7	+	Radial
Radial wrist extensor	Extensor carpi radialis longus	+	C6		Radial
Finger extensors	Extensor digitorum	+	C7	(+)	Posterior interosseous
Finger flexors	Flexor pollicis longus and flexor digitorum profundus, index Flexor digitorum profundus, ring and little		C8	+	Anterior interosseous Ulnar
Finger abduction	First dorsal interosseous Abductor pollicis brevis	++	T1 T1		Ulnar Median

The muscles listed in the 'Upper motor neurone' column are those that are preferentially affected in upper motor neurone lesions. The root level is the principal supply to a muscle.

In the acute injury, the object of the clinician must be to recognize the fact of injury as soon as possible after the event, and later, to go on to determine the nerve or nerves affected, the level(s) of injury, and the extent and depth of the lesion(s). The history is important: high injury transfer, open fracture and wounding (be it accidental, criminal, surgical, or all three) indicate a serious lesion and advice from witnesses or emergency paramedical staff is always valuable. Injuries that are potentially life- or limb-threatening complicate closed traction lesions of the supraclavicular brachial plexus in about 20% of cases. It is important to seek occult injuries to the head, spine, thorax, abdomen and pelvis before embarking on treatment of the nerve lesion(s).

The early symptoms of acute nerve injury include: abnormal spontaneous sensations; alteration or loss of sensibility; weakness and paralysis; impairment of function; an awareness of warming and dryness of all or part of an extremity (sometimes); and pain (sometimes). Neuropathic pain is never easy to recognize in an injured patient, who is probably confused, distressed and in pain. It can be distinguished from the pain of fracture or dislocation by loss of sensation; painful, spontaneous sensory symptoms, expressed throughout the territory of the nerve; and by lancinating or shooting pain, irradiating into the distribution of the nerve. Constant crushing, bursting or burning pain in an otherwise undamaged hand indicates serious and continuing injury to major trunk nerves more proximally. Progression of sensory loss, with a deep bursting or crushing pain within the muscles of the limb, signifies critical ischaemia.

The main nerves are stretched, ruptured or avulsed from muscle by violent hyperextension–abduction injuries of the arm. The axillary or brachial artery is ruptured in at least one-third of fractures of the proximal humerus or fracture dislocations of the shoulder. The axillary nerve and posterior circumflex vessels, which pass almost horizontally posterior to the neck of the humerus, are particularly at risk from anterior dislocation of the head of the humerus. The radial nerve and the profunda brachii artery are relatively fixed where they pass from the anterior to the posterior compartment by perforating the medial intermuscular septum in the axilla, and again when they re-emerge into the anterior compartment by perforating the lateral intermuscular septum. They may be stretched, ruptured or entrapped by displaced fractures of

the shaft of the humerus. The median, ulnar and radial nerves may be subjected to traction, or entrapment, by fracture or dislocation about the elbow. The median nerve is severely compressed by fractures or dislocations impinging into the carpal tunnel. All nerves may be transected by wounds from a knife, glass, a bullet or other missile anywhere along their course. The axillary artery and its main branches are commonly involved in nerve injuries. Wounds of the median nerve in the arm commonly extend to the brachial artery, and the ulnar artery is usually involved in wounds to the ulnar nerve in the forearm and at the wrist.

Nerves and related structures may be displaced by fracture, dislocation or haematoma. The ensuing fibrosis distorts normal anatomical relations; this increases the risk of intraoperative damage when nerves and accompanying vessels are not formally exposed. The incidence of such iatrogenic injuries is increasing, accounting for one-quarter of lesions of the radial nerve; the incidence is even higher for nerves around the elbow.

Injuries to nerves of cutaneous sensation are common after wounds, incisions or vessel puncture. It is remarkable that the elective removal of nerves such as the medial cutaneous nerve of the forearm for the purpose of grafting is followed by pain only very rarely, while lesions of the terminal branches are usually painful. The medial cutaneous nerve of the arm and the intercostobrachial nerve are at risk during operations in the axilla. The posterior cutaneous nerves of the arm and forearm are vulnerable during the lateral approach to the shaft of the humerus. The medial and lateral cutaneous nerves of the forearm are frequently damaged during operations around the elbow. The dorsal branch of the ulnar nerve is at risk during operations on the distal third of the ulna. The superficial radial and lateral cutaneous nerves are frequently injured by interventions on the lateral side of the wrist. Injuries to the palmar cutaneous branch of the median nerve and to the digital nerves are unwelcome complications of operations on the hand. These injuries cause pain, which may become severe. Mechanical allodynia, the perception of light touch as pain, is common; the spread of pain and allodynia beyond the territory of an injured nerve is frequent and indicates central sensitization of other neurones in the dorsal horn of the spinal cord.

EXAMINATION

Examination complements the history and the interpretation of symptoms to provide an accurate diagnosis. All findings should be set down in an intelligible record. Unfortunately, the signs of acute nerve injury have to be sought at a time when the patient may be least able to cooperate in an examination: soon after wounding, when there is likely to be distress and when the general condition may be affected by loss of blood and other injuries and/or by drink and drugs. The examination often has to be undertaken in the frequently unfavourable surroundings of an accident department. When the lesion has been inflicted by a surgeon or anaesthetist, the patient's response is likely to be distorted by postoperative pain, by the effects of recent general anaesthesia or by sedative or analgesic drugs. These are inappropriate conditions for a quiet and comprehensive 'neurological examination', and yet this is the time when the fact of nerve injury must at least be recognized if the best result is to be obtained from treatment. The examiner should, at all times, bear in mind the fact that, if there is a wound over the line of a main nerve and if there is any suggestion of loss of sensibility or impairment of motor function in the distribution of that nerve, it must be regarded as having been cut until and unless it is proved otherwise.

Local signs in closed injury

In closed injuries, swelling or ecchymosis and linear abrasions indicate severe soft tissue injury and also the line of application of force (see Fig. 6.1.1 in Commentary 6.1). Peripheral pulses and limb perfusion must be examined and re-examined.

Wound characteristics

It is useful to distinguish between the tidy wound caused by a knife and the untidy, and usually contaminated, wound of open fracture, machinery or gunshot. A further distinction is made between wounds caused by a handgun, rifle and close-range shotgun, the latter being particularly destructive. The injuries of current conflicts are classed as: penetrating missile wound, (small) fragment wound, and blast wound, those from mines or improvised explosive devices (IEDs).

Sensation

Sensory loss is determined by response to light touch and pinprick. If circumstances permit, the patient outlines the area of sensory loss, which is then marked by a black skin-marker pen. The surrounding zone of incomplete sensory loss can be similarly marked in red, and the limb then photographed. For clinical examples, see Figures 46.17–46.27.

Examination of muscles

Selected muscles are examined. The patient lying supine is usually able to demonstrate activity in serratus anterior by lifting the shoulders away from the couch, by means of 'forward shrugging'. It is usually possible to observe the presence of flexion and abduction at the shoulder, flexion



Fig. 46.17 The area of sensory abnormality after transection of C8 and T1. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.18 The area of sensory abnormality after section of the lateral cord in the axilla. As in Figure 46.17, there was no complete loss of sensation. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.19 Rupture of the posterior divisions of the trunks of the brachial plexus deep to the clavicle. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.20 Rupture of the axillary nerve. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.21 Sensory loss in two cases of rupture of the musculocutaneous nerve. Both patients were able to supinate the forearm fully; the power of elbow flexion was around 30%. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.22 A–B, Two examples of high lesion of the radial nerve. In **B**, there is early recovery into the wrist extensors after repair. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

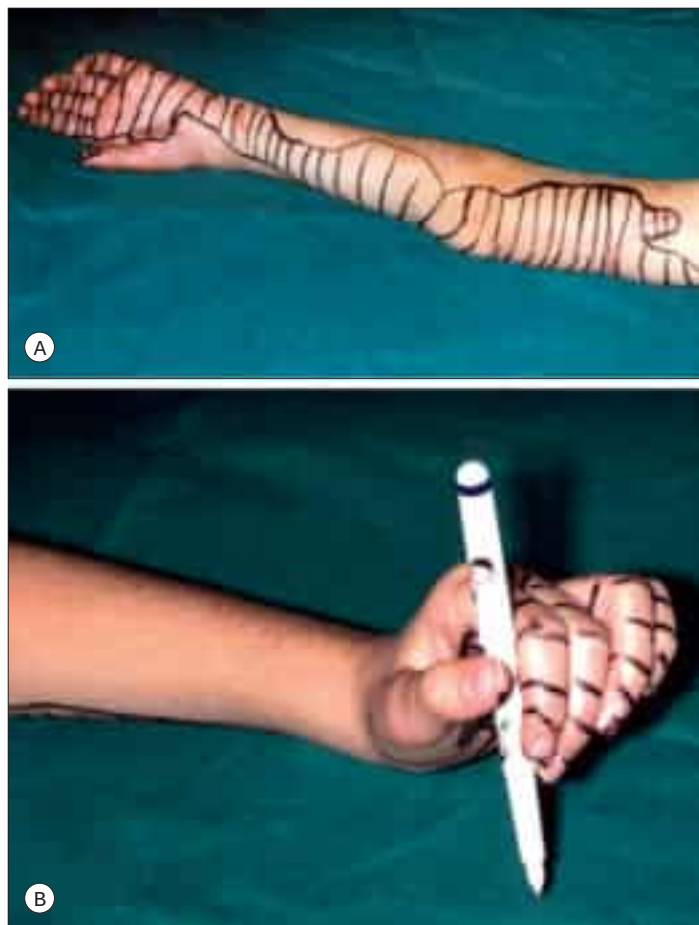


Fig. 46.23 A–B, Transection of the median nerve, ulnar nerve, medial cutaneous nerve of the forearm, and brachial artery, in the arm of a 14-year-old boy. Note the extent of skin innervation provided by the superficial radial and lateral cutaneous nerves of the forearm. The intact radial nerve permits a sort of grasp. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.24 A high median nerve injury; there is no active flexion of the index finger and thumb. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)



Fig. 46.25 Section of the median nerve and the palmar cutaneous nerve at the wrist. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

and extension of the elbow and wrist, and flexion and extension of the fingers. The radial, median and ulnar nerves are tested by asking the patient to form an 'O' between the thumb and little finger, to give the 'thumbs up' sign, and to open and close the fingers like a fan. The palmar skin is scrutinized for changes in colour and in sweating ([Fig. 46.28](#)). Although this may be more difficult in pigmented skin, such changes are detectable. The standard tendon reflexes are examined.

Secondary examination

A more detailed examination is possible when the patient's condition is stable, and when pain has been controlled. Limb dominance, occupation, marital status, underlying disease and continuing medication are recorded, if this has not already been done. Neuropathic pain is now somewhat easier to recognize because it is less responsive to analgesics than is pain from skeletal injury.

Recognition of the level of a lesion and the depth of injury

A sound grasp of the distribution of the spinal nerves and of the location of the formation and branching of the trunk nerves allows accurate diagnosis of the level of a lesion.

Posterior cord and radial nerve

The extent of the sensory loss is variable. There is no cutaneous sensory loss in lesions confined to the posterior interosseous nerve. The examination of selected muscles is reliable in guiding the clinician to the level of injury. They include, in order: teres major (inferior scapular nerve), latissimus dorsi (thoracodorsal nerve) and deltoid (axillary nerve). Subscapularis (superior scapular nerve) is not an easy muscle to examine.

Paralysis of all three heads of triceps places the injury in the axilla. The nerves to the lateral head of triceps leaves the main nerve rather distally, so that radial palsy, with paralysis of the lateral head but active medial and long heads, points to a lesion in the spiral groove. The nerve to brachioradialis consistently passes away from the main trunk about three fingers' breadths above the lateral epicondyle, and the nerve to extensor carpi radialis longus leaves the main trunk about 1 cm more distally. One nerve to extensor carpi radialis brevis leaves the main nerve about 1 cm proximal to the lateral epicondyle, and another leaves at the site of division into the superficial radial and posterior interosseous nerves.

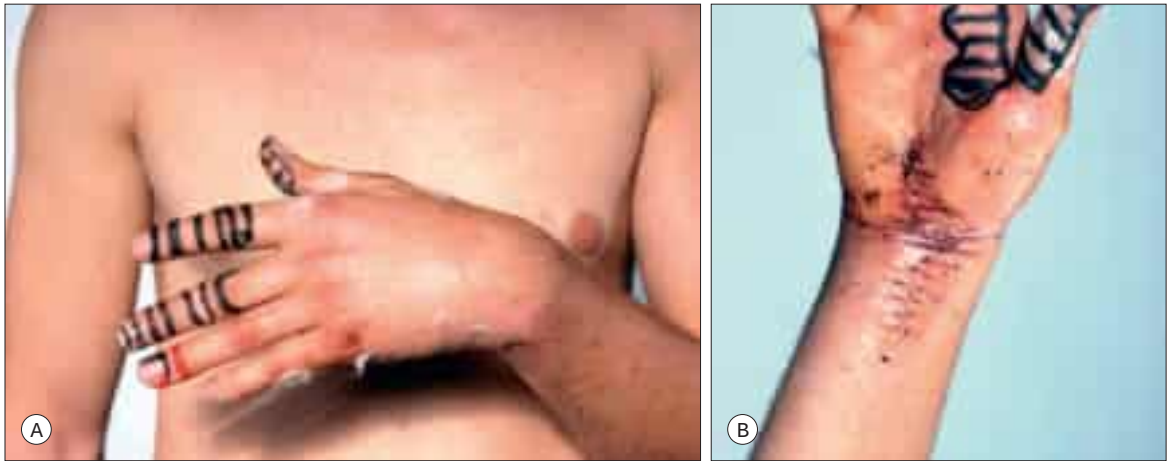


Fig. 46.26 A–B, A typical area of loss of sensibility after division of the median nerve at the wrist, sparing the palmar cutaneous branch. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

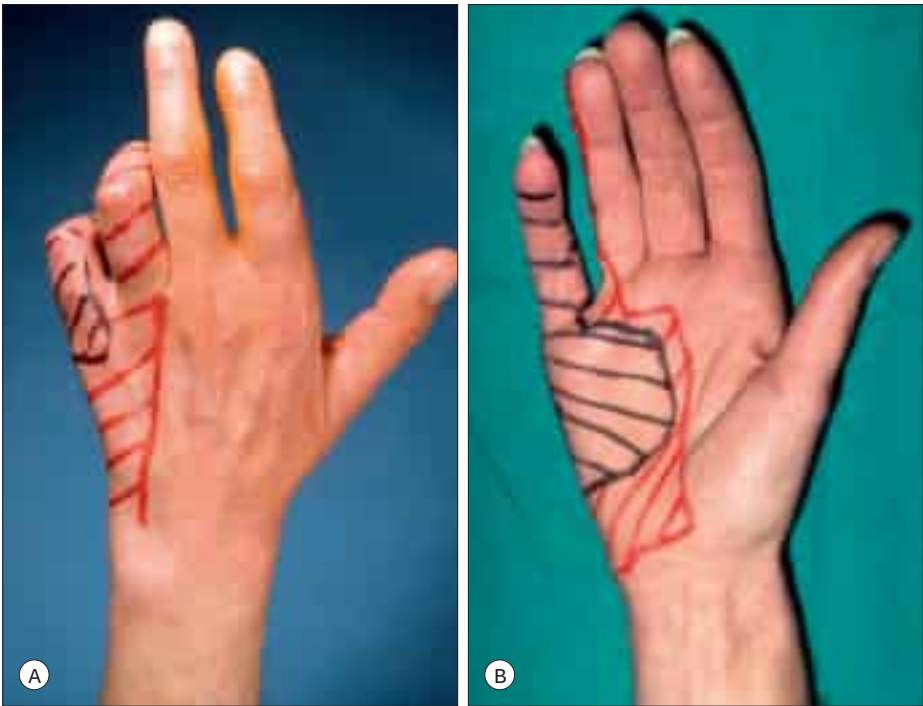


Fig. 46.27 The area of sensory loss and ‘clawing’ of little and ring fingers after division of the ulnar nerve in the forearm. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

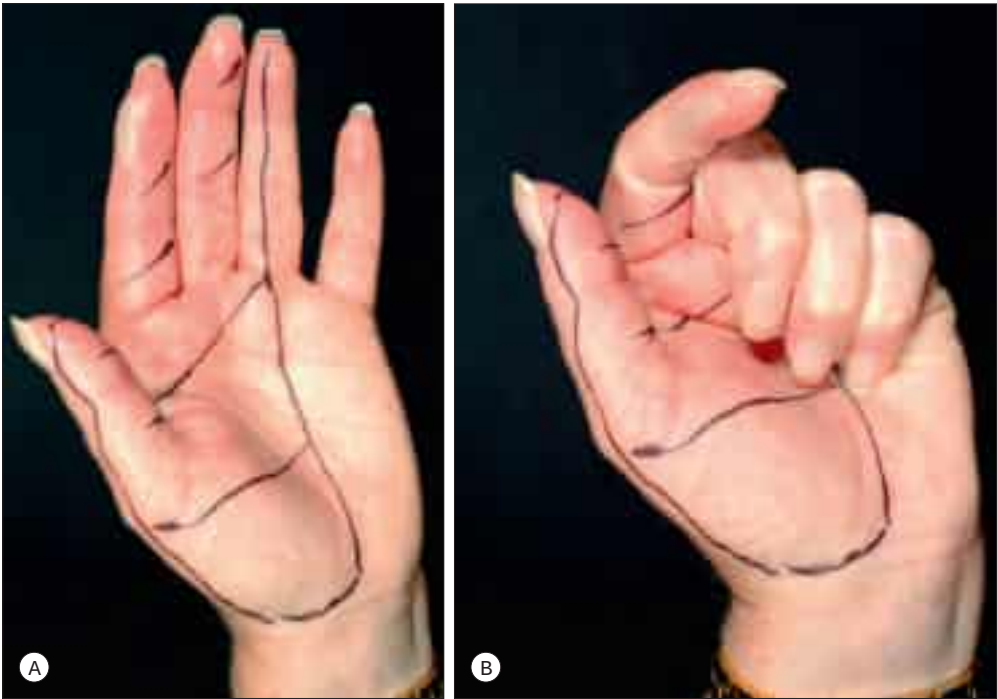


Fig. 46.28 Sympathetic paralysis seen within a few days of transection. **A, B,** Transection of the median nerve at the elbow. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

Lateral cord, musculocutaneous nerve and lateral root of the median nerve

It seems that most of the median nerve fibres that innervate the skin of the hand pass through the lateral cord. Paralysis involves biceps brachii, brachialis, flexor carpi radialis and pronator teres. That portion of brachialis innervated by the radial nerve may be intact. Division of the musculocutaneous nerve does not lead to loss of elbow flexion because brachioradialis acts as a powerful flexor of the joint (see Fig. 46.21).

Medial cord, medial head of the median and ulnar nerves

The sensory loss does not extend into the arm. If it does, then the medial cutaneous nerve of the arm must be involved. Paralysis is extensive, and includes the long flexor muscles to the digits, the small muscles of the hand, and the smooth muscle associated with the vessels and sweat glands supplying the ulnar three fingers.

Median nerve

Involvement of the palmar cutaneous nerve is useful in localizing a lesion at the wrist (see Figs 46.25, 46.26), but the extent of paralysis is a more precise guide to the level of a lesion. Injuries in the arm cause paralysis of all muscles innervated by the median nerve, including those arising from the medial epicondyle (flexor carpi radialis, pronator teres, palmaris longus, and part of flexor digitorum superficialis), as well as those innervated by the anterior interosseous nerve (flexor pollicis longus, flexor digitorum profundus to index finger) (see Fig. 46.24). Anterior interosseous palsy is distinguished from high median injury by the absence of any loss of skin sensation. Abductor pollicis brevis is the one small muscle of the hand that is consistently innervated by the median nerve.

Ulnar nerve

The sensory loss is related to the involvement of the dorsal and palmar cutaneous branches (see Fig. 46.27). Ulnar palsy with paralysis of flexor carpi ulnaris places a lesion in the arm. Paralysis confined to the hand places a lesion in the distal half of the forearm. Preservation of the hypothenar muscles with paralysis of the other hand muscles innervated by the ulnar nerve places a lesion in the proximal part of the palm of the hand. There is no cutaneous sensory loss in a palsy of the deep branch of the ulnar nerve. The posture of the digits is characteristic in ulnar nerve lesions.

The examiner should bear in mind the possibility of anatomical variation and also the ability for other muscles to compensate for the actions of those paralysed by what are termed 'trick movements'. In fact, these muscles are acting appropriately and the movements are real enough (see Fig. 46.23B).

Depth of a lesion

The distinction between conduction block and severance of an axon (axonotmesis or neurotmesis) is sometimes difficult. Serious injuries are likely to cause serious lesions of nerves. Severance of a nerve with a cutaneous sensory component will lead to well-defined loss of sensibility and to complete motor, sudomotor and vasomotor paralysis in the distribution of the nerve. Simple conduction block is likely to produce a patchy loss of sensibility and a patchy motor loss. Further, it is likely to bear more heavily on the large axons than on the small ones;

vibration sense and sensibility to light touch are likely to be impaired, whereas pain sensibility may be unaffected. There is a risk of underestimating the severity of a nerve lesion in cases of closed injury and after intraoperative events. Different populations of nerve fibres within a main nerve that has been violently stretched but not ruptured sustain lesions of all grades of severity. Some remain intact, whilst conduction is blocked in others. Some ruptured axons will recover spontaneously, whilst others will not, depending on their cellular microenvironment (see Commentary 1.6).

Testing of sensibility is often difficult soon after wounding, or when nerve injury is associated with fracture of a long bone. The actions of some muscles can be simulated by the actions of others, so that the fact of paralysis can be missed in the early stages after nerve injury. Thus, when the median nerve has been cut, opposition of the thumb can be mimicked by the action of extensor pollicis brevis (ulnar nerve) and abductor pollicis longus (radial nerve). Similarly, abductor pollicis longus can imitate the action of the abductor pollicis brevis. It is important to examine both hands, especially in the early days, when there is no wasting to guide the examiner. One almost infallible sign is always present in the first 48 hours after deep injury of a nerve with a cutaneous sensory component: because small, as well as large, fibres are affected, the skin in the distribution of the damaged nerve is warm and dry. In a small child, there may be an abnormal posture of the denervated digits; the 'immersion test' is also helpful. The injured hand or foot is placed, for a few minutes, in warm water; the skin of the denervated digits fails to wrinkle. In ischaemia, the large fibres conveying discriminative sensibility and vibration sense are first affected. It is not easy to test these modalities when ischaemia is developing because of the damage to a main vessel that is associated with a fracture of a long bone, but if action is not taken until superficial sensibility is lost, it will come too late.

Tinel's sign

In closed injuries, percussion of the skin over a nerve in which axons have been ruptured evokes sensations usually described as a wave or surge of pins and needles into the cutaneous distribution of the nerve. This is Tinel's sign and it is a most useful aid to diagnosis. The sign is elicited on the day of injury in most conscious patients. It indicates not only where the nerve has been injured but also the fact that at least some axons have been ruptured. Tinel's sign can be detected from such 'motor' nerves as the posterior interosseous nerve; sensory symptoms radiate into muscular territory rather than into skin. It is more difficult to elicit the sign over deep-seated nerves such as the axillary, eighth cervical or first thoracic nerves.

Eliciting Tinel's sign in closed lesions

The examiner's finger percusses along the course of the nerve from distal to proximal, starting below the presumed level of lesion. The patient is asked to say when the advancing finger elicits a wave or a surge of pins and needles or abnormal sensations, which may be painful, into the distribution of the nerve, which must be clearly indicated by the examiner. The level of the sign should be measured from a fixed bony point and the distance recorded. At times, the examination is painful and patients need to be warned of this. A Tinel's sign that remains static at the level of a lesion strongly suggests rupture of the nerve or persisting local conditions inimical to spontaneous regeneration.

14

Common examples of the classical compartment syndrome in the upper limb occur in the flexor and extensor compartments of the forearm. Although nerve conduction is lost before muscle contractility, ischaemia is more likely to produce infarction of muscle than necrosis of nerve. Decompression of a nerve embedded within muscle that has become fibrosed by ischaemia usually leads to relief of pain and considerable improvement in sensation and sympathetic function, even when delayed for some months.

Anoxia occurs either because of cessation of flow through a main axial artery or as a result of increasing pressure within an osseofascial compartment caused by bleeding, infusion of fluid or sepsis. It destroys the integrity of the cell membrane and capillary endothelium, which means that the homeostatic balance between the intravascular, extracellular and intracellular spaces is lost. Tissue death is imminent.

The increased permeability of the vascular endothelium leads to an increase in intracompartmental pressure by exudation, which collapses the low-pressure lymphatic and venous systems. The final event is the closure or obstruction of perfusing arterioles to muscles and the extrinsic supply to nerves, when tissue or extravascular pressure exceeds the cortical closing pressure of those vessels. The vicious circle is complete. The fluid entering the compartment cannot get out and continues to leak into the compartment until the pressure is so high that inflow is blocked. Increasing intracompartmental pressure occurs when the collateral circulation is inadequate after occlusion of the main artery or after restoration of flow through that vessel, unless the compartments have been decompressed adequately by fasciotomy. The vascular anastomoses about the joints of the upper limb provide a richer collateral circulation than those found in the lower limb. In a study of nearly 2,500 World War II battle casualties, at a time when ligation of damaged arteries was common, ligation of the popliteal artery led to amputation in 346 of 502 limbs. In contrast, the incidence of amputation after ligation was about one-quarter for the subclavian artery, one-third for the axillary artery and more than one-half when the brachial artery was ligated proximal to the profunda brachii artery. These figures are certainly lower in the less contaminated injuries of civilian life, but some degree of post-ischaemic fibrosis is almost inevitable after failure to repair these vessels. The principles of arterial repair include urgent restoration of flow by repair of the artery and restoration of tissue perfusion by decompression. Temporary intraluminal shunts are invaluable, as they buy time and permit adequate stabilization of the skeleton. Application of these principles in current conflicts ensures a remarkably high rate of limb survival and a remarkably low incidence of post-ischaemic fibrosis.

Flow through the brachial artery in adults and in children has been measured by high-resolution ultrasonography. Flow was calculated by the Laplace equation: $BF(\text{blood flow}) = [\pi \times (D/2)]^2 \times FV(\text{flow velocity})$. The mean diameter of the brachial artery in children aged between 4 and 5 years is 2.7 mm, which provides a resting flow of about 200 ml per minute. The significance of the diameter of the vessel is emphasized by Poiseuille's law. This is the physical law describing the volume of flow (Φ) of an incompressible uniform viscous liquid, where R is the internal radius of the tube, P the pressure difference between the two ends, η the dynamic fluid viscosity and L the total length of the tube.

$$\Phi = \frac{\pi R^4}{8\eta} \frac{\Delta P}{L}$$

The diameter of the superior ulnar collateral artery at the elbow in a 5-year-old child is, at most, 1 mm. This calibre provides flow of about 20 ml per minute, assuming that the pressure gradient is the same as that in the brachial artery itself. These facts must be borne in mind by any clinician inclined to the view that cessation of flow through the brachial artery is a matter of little consequence. The collateral channels take hours or days to develop and may be compromised by the original injury or during operation. The radial and ulnar nerves are often entrapped or compressed in fracture dislocations at the elbow, blocking flow through the ulnar and radial collateral systems that accompany these nerves.

The presence of peripheral pulses does not indicate adequate perfusion of the deep intracompartmental tissues; neither does the perfusion of skin or the nail bed when pulses are absent. The cardinal symptom of ischaemic anoxia is severe pain and the cardinal sign is loss of nerve conduction. This can be confirmed during exposure of limb nerves with an inflated tourniquet cuff in position proximally. For about 20 minutes, stimulation of the nerve evokes a brisk muscular response as a consequence of transmission through the neuromuscular junction. This response diminishes and disappears after

about 30 minutes, although conduction within the nerve itself can still be detected for up to another 20 minutes. However, direct stimulation of the muscle provokes a twitch that can be elicited for up to several hours. Indeed, it is the loss of this direct response that signifies impending death of the muscle, and with it, death of the limb. The earliest nerve fibres to suffer are the largest, i.e. those conveying vibration and proprioception sense.

Ischaemia and acute compression within neurovascular fascial compartments

Sleeves of fascia surround main nerves and main vessels in some regions, an arrangement that predisposes nerves to injury from ischaemia or compression, or both. The ventral (anterior) primary rami of C7, C8 and T1 are enclosed in quite a rigid space after they enter the posterior triangle of the neck. This is bounded, posteriorly, by the dorsal part of the first rib, the transverse processes of the cervical vertebrae and by the fascia of levator scapulae. The nerves are embraced by scalenus anterior and scalenus medius, both of which are invested in an unyielding fascia. This is one envelope of the prevertebral fascia that also serves to bind the phrenic nerve down to the anterior face of scalenus anterior. The prevertebral fascia is particularly well developed in front of the vertebral column and also at the base of the posterior triangle, where it envelops C7, C8 and T1, the phrenic nerve, the cervical sympathetic chain, and the subclavian and vertebral arteries. Infusion of relatively large volumes of fluid, from 10 to 20 ml, deep to the prevertebral fascia for the purpose of inducing regional block may cause tamponade of the radicular vessels that enter the spinal canal and contribute to the anterior spinal artery. Wilbourn (2005) has described the medial brachial fascial compartment, which extends from the axilla to the elbow and is bounded by the tough medial intramuscular septum and the axillary sheath. Bleeding into this compartment is responsible for the majority of infraclavicular plexopathies that follow regional block, and for many of the neurological lesions that result from closed or even open injuries in this region. The anterior interosseous nerve and its accompanying artery may be damaged by compression because of swelling in the deepest part of the flexor compartment of the forearm. The ulnar nerve, accompanied by the ulnar artery, lies in a discrete fascial compartment in the distal two-thirds of the forearm.

Bleeding into the axillary fascia (the medial antebrachial compartment) causes a characteristic, progressive lesion. There is, almost always, pain accompanied by dysaesthesiae; loss of sensation soon follows and then, over the next 2–3 hours, paralysis ensues. Wilbourn's comment bears repeating:

Distal pulses are normal as they are with most compartment syndromes because the elevated pressure, although sufficient to collapse the vasa nervorum, is far below mean arterial pressure. Ultrasound, MR and CT may reveal the vascular lesion, but, considering the very brief time available for surgical decompression before irreversible nerve damage occurs, obtaining these is rarely justified.

In one series of 16 patients (Stenning et al 2005), there was, in all cases, an injury to the axillary artery or one of its offsets, caused by dislocation of the shoulder or fracture of the proximal humerus. The diagnosis of continuing bleeding into the axillary sheath was made by the delayed onset of nerve palsy or the deepening of the lesion whilst under observation. A favourable outcome was seen in 87 nerve palsies, where urgent repair of the artery and decompression of the axillary sheath were performed. However, delay in diagnosis and treatment may lead to permanent loss of function (see Fig. 6.2.17).

Case report After a fracture/dislocation of the shoulder in a 59-year-old man, the patient's complaints of pain and the signs of a deepening lesion were not appreciated for several days. Angiography showed bleeding from the subscapular artery. Two unsuccessful attempts were made to occlude this by embolization. The patient was reviewed at 6 weeks in right heart failure; haemoglobin was 4.9 g/litre (49 g/dl). He was in great pain and there was a total and deep plexopathy. A magnetic resonance imaging (MRI) scan showed an enormous haematoma occupying the axilla and the arm, and suggested continuing bleeding. Treatment was simple: the axillary artery was exposed and controlled above pectoralis minor and the brachial artery exposed and controlled in the arm. More than 4 litres of altered blood were removed from the sac. The defect in the axillary artery was, at most, 2 mm in diameter and it was closed by direct suture. The pain was relieved and there was gradual recovery of the nerves, but the small muscles of the hand never recovered.

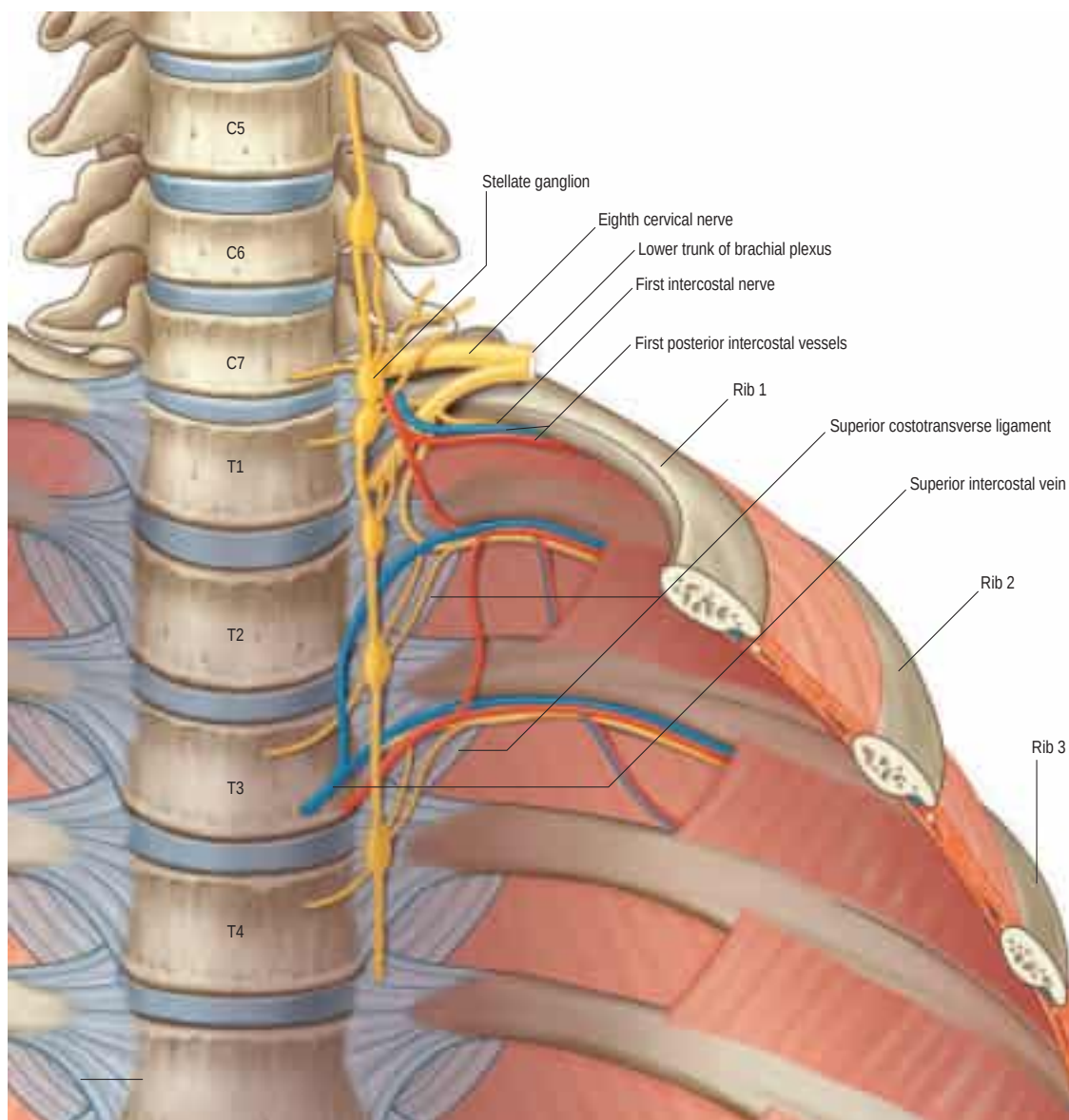


Fig. 46.16 The relations of the left cervicothoracic (stellate) ganglion. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

The infraclavicular fossa (deltopectoral triangle) is visible as a small depression located inferior to the clavicle between the clavicular origins of pectoralis major and deltoid, at the junction of the convex medial and concave lateral portions of the clavicle (see Fig. 46.29A). The apex of the coracoid process lies approximately 2.5 cm below the clavicle immediately lateral to this fossa, and is covered by the anterior fibres of deltoid. Deep palpation lateral to the coracoid process and through deltoid reveals the lesser tubercle of the humerus below the tip of the acromion. The greater tubercle of the humerus is the most lateral bony point in the shoulder region and is palpable inferolateral to the acromion. The greater and lesser tubercles can be felt moving under an examining finger during lateral and medial humeral rotation. The vertical groove or indentation of the intertubercular (bicipital) groove is palpable between the tubercles; regional tenderness occurs in bicipital tendinitis. When the arm is abducted, the head of the humerus can be palpated on deep pressure in the apex of the axilla.

The shaft of the humerus is covered by overlying muscles and is not easily palpable. Distally, the medial humeral epicondyle is both visible and palpable, particularly when the elbow is flexed, and can be traced proximally to the medial supracondylar ridge. The ulnar nerve passes posterior to the medial epicondyle and can be rolled from side to side just proximal to the epicondyle (Fig. 46.29B). The lateral epicondyle is less prominent, but its posterior surface is palpable and its lateral margin can be traced proximally on deep palpation to the lateral supracondylar ridge. Inspection of the posterior surface of an extended elbow reveals a depression lateral to the midline, bounded laterally by the fleshy elevation formed by the superficial forearm extensor muscles, and medially by the lateral side of the olecranon. The floor of this depression contains the posterior surface of the lateral epicondyle superiorly,

and the head of the radius covered by the anular ligament inferiorly. Rotations of the radial head are palpable during forearm pronation and supination. The humero-radial part of the elbow joint forms a palpable transverse depression between the lateral epicondyle and the radial head.

The posterior surface of the olecranon is subcutaneous and tapers as it passes inferiorly. When the elbow is extended, the apex of the olecranon lies on a line level with the medial and lateral humeral epicondyles (see Fig. 46.29B). When the elbow is flexed, the apex of the olecranon descends and the three bony points then form a triangle. This relationship is lost following elbow dislocation.

The elbow joint is situated 2 cm below a line joining the two humeral epicondyles. It slopes downwards and medially from its lateral extremity. When the elbow is fully extended with the forearm and hand in supination, the carrying angle is normally 165° in females and 175° in males. The angle is obscured in full forearm pronation and disappears on full elbow flexion, when the shafts of the ulna and humerus come to lie in the same plane. The elbow joint can be accessed with a needle from a posterior perspective *via* the anconeus triangle; the borders of the triangle join the olecranon, radial head and lateral epicondyle.

The posterior border of the ulna is subcutaneous and palpable from the olecranon superiorly to the styloid process inferiorly (see Fig. 46.29C). Its position corresponds to the longitudinal furrow that is visible on the posterior forearm when the elbow is fully flexed and marks the border between the anterior and posterior forearm compartments. In contrast, the radial shaft is covered by muscles and can only be felt indistinctly. The rounded head of the ulna forms a surface elevation on the medial part of the posterior aspect of the wrist (Fig. 46.30B),

especially when the hand is pronated. The ulnar styloid process projects distally from the posteromedial aspect of the ulnar head.

The expanded distal end of the radius forms a slight surface elevation on the lateral side of the wrist and can be traced downwards to the radial styloid process. The posterior aspect of the lower end of the radius is partly obscured by the extensor tendons but can be palpated without difficulty. The prominent dorsal radial (Lister's) tubercle is both visible and palpable, and aligns with the web space between digits 2 and 3 (Fig. 46.30B–C). The tendon of extensor pollicis longus in extensor compartment 3 sits medial to the tubercle, and the tendons of extensor carpi radialis longus and brevis sit lateral to the tubercle in compartment 2.

The wrist joint is identified between the distal ends of the radius and ulna and the proximal carpus on flexion and extension of the wrist. The line of the wrist joint corresponds to a line, convex superiorly, joining the radial and ulnar styloid processes. Its position is also marked by the proximal of the two transverse anterior wrist creases (Fig. 46.30A). The radial styloid process normally sits 1 cm distal to the ulnar styloid process. Alteration of this arrangement can indicate fracture.

Four of the carpal bones are palpable. The pisiform is both visible and palpable on the palmar aspect of the medial wrist at the base of the hypothenar eminence. It is often overlain by the distal wrist crease.

It can be moved over the articular surface of the triquetrum when the wrist is passively flexed. The hook of the hamate lies approximately 2.5 cm distal to the pisiform, in line with the ulnar border of the ring finger. It can be felt on deep palpation, and the superficial division of the ulnar nerve can be rolled from side to side over the tip of the hook. The tubercle of the scaphoid is situated at the base of the thenar eminence, in line with the tendon of flexor carpi radialis, and in many individuals forms a small visible elevation. Immediately distal to it, but covered by the muscles of the thenar eminence, the tubercle of the trapezium can be identified on deep palpation.

The dorsal aspects of the trapezium and of each of the metacarpal bones are partially palpable around the margins of the corresponding extensor tendons (see Fig. 46.30B). The heads of the metacarpal bones form the prominences of the knuckles, and are most obvious during digit flexion. On the dorsal hand, the metacarpophalangeal joint line forms a palpable depression lateral to the digital extensor tendon, and alternating digit flexion/extension aids identification. The joint can be aspirated/injected from this perspective. Palpation distal to the metacarpal head reveals the flared base of the corresponding proximal phalanx. The interphalangeal joints are palpable on the dorsal aspect of a flexed digit just distal to the prominences formed by the heads of the proximal and middle phalanges.

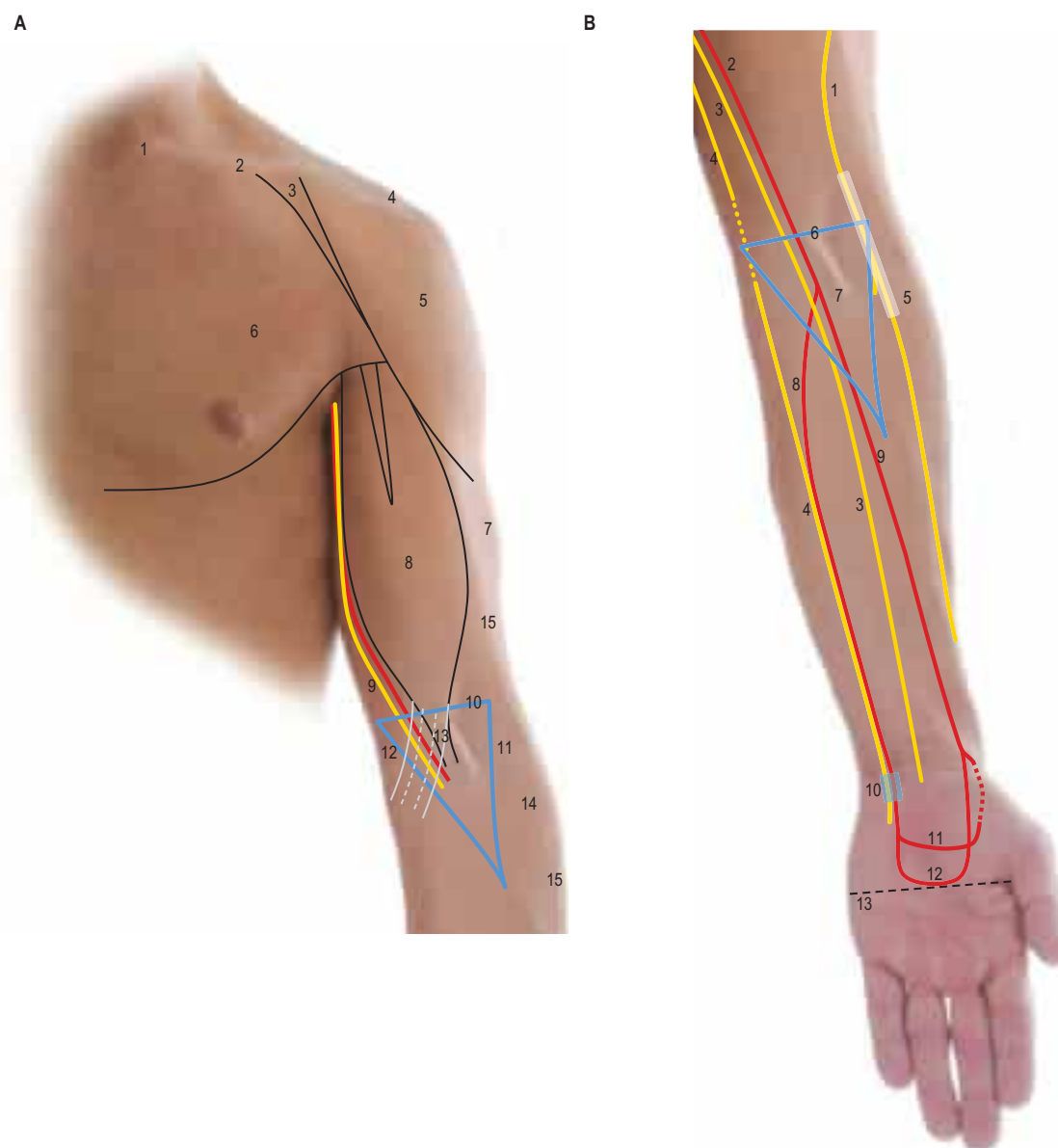


Fig. 46.29 Surface anatomy of the upper limb. **A**, The anterior aspect of the left pectoral girdle, arm and elbow. Key: 1, sternoclavicular joint; 2, clavicle; 3, infraclavicular fossa/deltopectoral triangle, and deltopectoral groove running inferolaterally; 4, acromion; 5, deltoid; 6, pectoralis major; 7, lateral head of triceps; 8, biceps brachii; 9, median nerve and brachial artery running anterior to triceps medial head; 10, epicondylar line and borders of cubital fossa (blue); 11, brachioradialis; 12, pronator teres; 13, biceps tendon (black) and aponeurosis (grey) passing inferomedially; 14, extensor muscle mass; 15, cephalic vein. **B**, The anterior aspect of the left elbow, forearm and hand. Key: 1, radial nerve; 2, brachial artery; 3, median nerve; 4, ulnar nerve (dashed part of line shows nerve passing posterior to medial epicondyle); 5, zone of radial nerve bifurcation (white): 1.0 ± 2.8 cm distal to the lateral epicondyle; 6, cubital fossa borders (blue); 7, biceps tendon with brachial artery bifurcation medially; 8, ulnar artery; 9, radial artery; 10, pisiform with Guyon's canal (blue) laterally; 11, deep palmar arch; 12, superficial palmar arch; 13, plane of fully extended thumb.

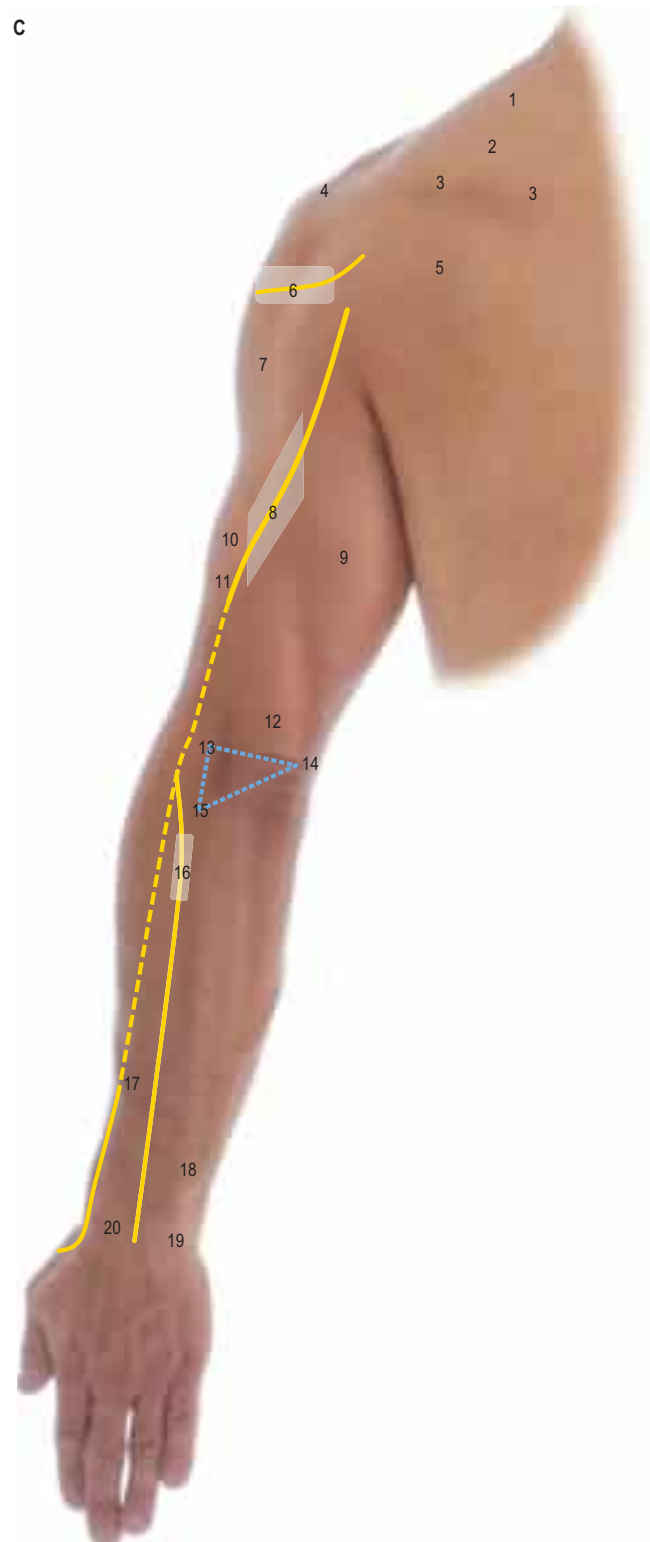


Fig. 46.29, cont'd C. A posterior view of the left pectoral girdle, arm, elbow and forearm. Key: 1, trapezius; 2, supraspinatus; 3, spine of scapula; 4, acromion; 5, infraspinatus; 6, zone of axillary nerve passing around proximal humerus (white): approximately 6.0 cm from proximal humeral margin; 7, deltoid; 8, radial nerve and its zone of passage around spiral groove (white): enters the spiral groove approximately 46% and leaves it approximately 60% of the way down a line from greater tubercle to lateral epicondyle; 9, triceps, long head; 10, triceps, lateral head; 11, radial nerve piercing intermuscular septum approximately 66% of the way down a line from greater tubercle to lateral epicondyle; 12, triceps tendon; 13, lateral epicondyle and borders of anconeus triangle shown as blue dashed line; 14, olecranon and olecranon bursa; 15, radial head; 16, zone of posterior interosseous nerve winding around radial neck (white): approximately 2.5 cm distal to radial head; 17, superficial radial nerve emerging from between the tendons of brachioradialis and extensor carpi radialis longus approximately 8.5 cm proximal to radial styloid process; 18, posterior border of ulna; 19, ulnar head; 20, dorsal radial tubercle.

The flexor retinaculum, and therefore the surface position of the carpal tunnel, can be mapped according to its palpable bony attachments (see Fig. 46.30A). The distal border is concave inferiorly and is marked by a curved line that joins the tubercle of the trapezium to the hook of the hamate. Its proximal border is marked by a curved line, concave superiorly, that joins the tubercle of the scaphoid to the pisiform. The dominant, most distal, wrist crease normally sits distal to the lunate and overlies pisiform and the proximal edge of the flexor retinaculum. The carpal tunnel passes distal to this crease for 2.5–3 cm.

MUSCULOTENDINOUS LANDMARKS

Deltoid can be identified when the arm is abducted against resistance. The deltoid tendon is palpable approximately halfway down the lateral aspect of the humerus. Its anterior border can be traced superomedially from the anterior aspect of the humerus, across the tendon of pectoralis major, to form the lateral boundary of the infraclavicular fossa (see Fig. 46.29A, Video 46.1). The posterior border runs superomedially from the posterior aspect of the deltoid tendon and reaches the crest of the scapular spine near its medial end. The normal rounded contour of the shoulder is produced by deltoid covering the lateral aspect of the greater tubercle of the humerus. Shoulder dislocation results in the loss of the normal rounded contour of the shoulder because the greater tubercle is displaced medially, and deltoid consequently descends vertically to its humeral attachment. Supraspinatus and infraspinatus are visible and palpable above and below the scapular spine, respectively (see Fig. 46.29C).

The lower border of pectoralis major forms the anterior axillary fold. It is visible and palpable when an abducted shoulder is adducted against resistance, e.g. by firmly pressing a hand on to the hip. The clavicular head of pectoralis major can be felt and seen to contract when flexing the shoulder to a right angle against resistance; the sternocostal head becomes visible when extending a flexed shoulder against resistance.

The posterior axillary fold, produced by latissimus dorsi and the underlying teres major, reaches a lower level on the humerus than the anterior axillary fold. When the abducted shoulder is adducted against resistance, the posterior axillary fold is accentuated and the lateral border of latissimus dorsi can be traced inferomedially to its attachment to the iliac crest. When the upper limb is raised above the head, the lower five or six serrations of serratus anterior are visible on the lateral aspect of the thorax; they pass downwards and forwards to interdigitate with the serrations of external oblique.

Biceps brachii produces a visible elevation on the anterior arm and shallow furrows mark its medial and lateral borders (see Fig. 46.29A). It diminishes superiorly where it is covered by pectoralis major, and inferiorly where it tapers to the biceps tendon just above the elbow joint. Flexion of the elbow against resistance aids identification of the muscle and of the bicipital tendon, which can be held between finger and thumb and traced down into the cubital fossa. With the elbow held in this position, the sharp upper margin of the bicipital aponeurosis can be traced passing inferomedially over the elevation produced by the superficial group of forearm flexor muscles. Coracobrachialis emerges from the lateral axillary wall and forms a rounded ridge on the upper part of the medial side of biceps.

On the posterior arm, the lateral head of triceps forms a laterally placed elevation, inferior to the posterior border of deltoid (see Fig. 46.29C). Triceps can be made visible by extending the elbow against resistance. On the medial side, the fleshy mass produced by the long head of triceps passes superiorly and disappears deep to deltoid.

Brachioradialis is the most superficial muscle on the lateral side of the forearm. When the elbow is flexed against resistance whilst in a mid pronation-supination position, brachioradialis stands out as a prominent ridge extending upwards beyond the level of the elbow joint to the lower lateral side of the arm. The medial (ulnar) border of brachioradialis marks the boundary between the anterior and posterior forearm compartments.

The three borders of the cubital fossa can be mapped out anterior to the elbow by a line joining the medial and lateral humeral epicondyles (superiorly), brachioradialis (laterally) and pronator teres (medially) (see Fig. 46.29A–B). Pronator teres passes inferolaterally from the medial epicondyle to a point approximately halfway down the radial shaft; its rounded belly is visible during resisted forearm pronation.

The fleshy mass of the superficial forearm flexors is visible and palpable in the upper medial forearm, and that of the extensor muscle mass is visible in the upper lateral forearm. The line of travel of the superficial forearm muscles can be mapped out according to their

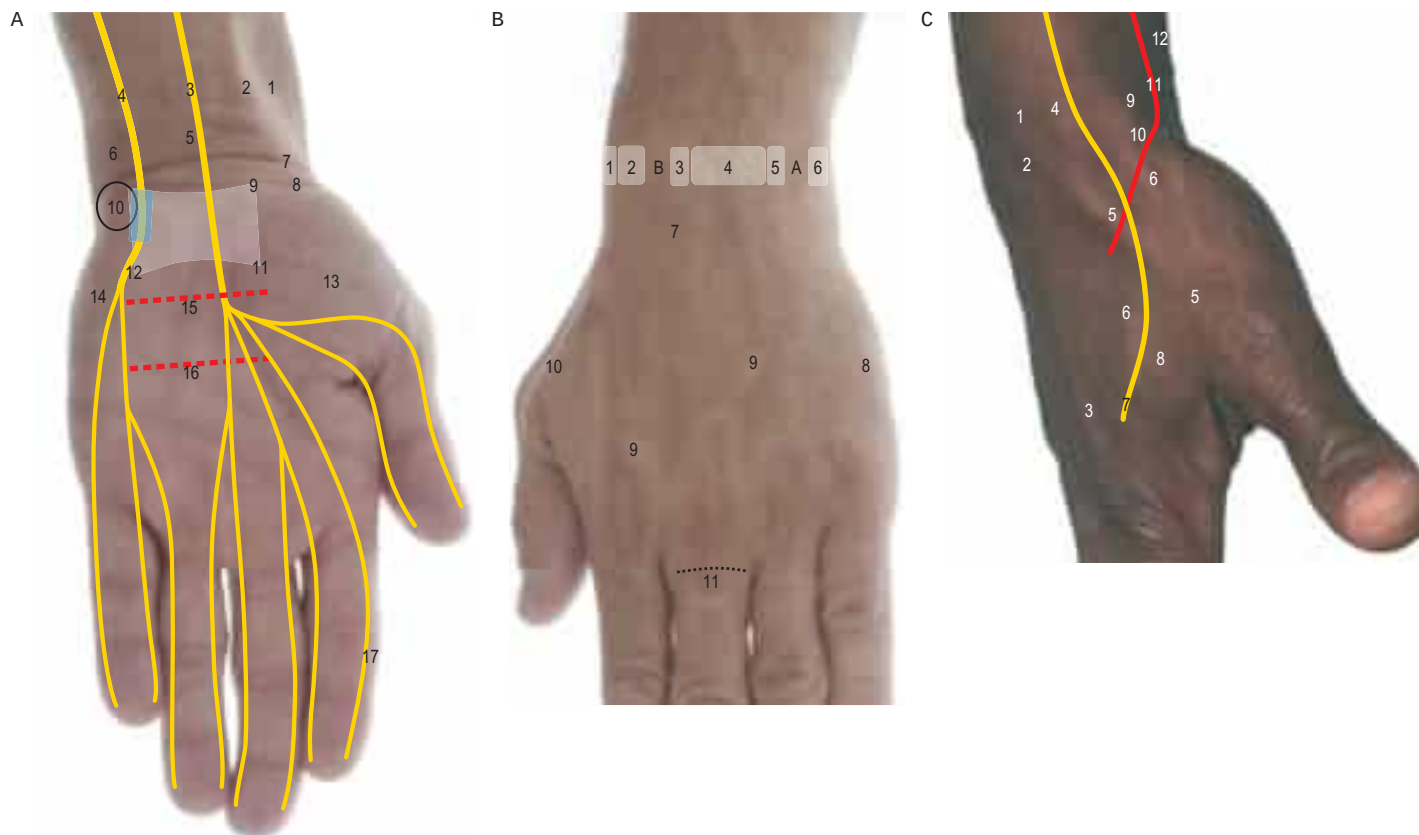


Fig. 46.30 The surface anatomy of the wrist and hand. **A**, Key: 1, radial artery; 2, flexor carpi radialis tendon; 3, median nerve; 4, ulnar nerve; 5, palmaris longus tendon; 6, flexor carpi ulnaris tendon; 7, proximal transverse wrist crease: marks the line of the wrist joint; 8, distal transverse wrist crease: marks the proximal boundary of the carpal tunnel; 9, scaphoid tubercle; 10, pisiform (black oval): aligns with distal wrist crease; 11, trapezium tubercle; 12, hook of hamate and superficial branch of ulnar nerve passing anteriorly; 13, thenar eminence; 14, hypothenar eminence; 15, plane of deep palmar arterial arch; 16, plane of superficial palmar arterial arch; 17, positions of palmar digital neurovascular bundles; shaded rectangle indicates position of flexor retinaculum. **B**, Key: 1–6, first to sixth extensor compartments of the wrist; 7, wrist joint needle insertion point; 8, hypothenar eminence; 9, dorsal venous network of the hand; 10, extensor pollicis longus tendon; 11, extensor tendon and metacarpophalangeal joint line; A, head of the ulna; B, dorsal radial (Lister's) tubercle. **C**, Key: 1, dorsal radial (Lister's) tubercle; 2, wrist joint needle insertion point; 3, extensor digitorum tendon; 4, extensor compartment 2, containing the tendons of extensors carpi radialis longus and brevis; 5, extensor pollicis longus tendon; 6, cephalic vein; 7, superficial branch of radial nerve; 8, first dorsal interosseous muscle; 9, extensor compartment 1, containing abductor pollicis longus and extensor pollicis brevis; 10, radial styloid process; 11, radial artery; 12, flexor carpi radialis tendon.

attachment points. Inflammation of the common flexor or extensor tendons can be indicated by tenderness to palpation over the respective muscle mass or humeral epicondyle.

Palpating lateral to flexor carpi radialis, 3–4 cm proximal to the wrist crease, reveals the muscle belly of flexor pollicis longus (flexing and extending the thumb will confirm that the examining finger is correctly placed). The area on the ulnar side of flexor carpi radialis tendon is packed with functionally important structures. The median nerve is either covered by, or situated just lateral (radial) to, the tendon of palmaris longus. The latter can be identified by gently pinching the tips of the digits together and partially flexing the wrist (see Fig. 46.30A). Near to the wrist, the median nerve lies very close to the skin and can, therefore, be injured by relatively superficial lacerations. When palmaris longus is absent, only a thin covering of subcutaneous fat and deep fascia separate skin and nerve. The four tendons of flexor digitorum superficialis lie deep to the median nerve; the tendons to the middle and ring fingers lie anterior to those for the index and little fingers as they pass deep to the flexor retinaculum, and can be felt and, usually, seen to move during finger flexion/extension. Deeper still are the tendons of flexor digitorum profundus. The broad tendon of flexor carpi ulnaris is easily palpated passing to the pisiform on the ulnar side of the anterior wrist; the ulnar nerve, artery and venae comitantes lie along its lateral (radial) edge. Any sharp injury that cuts through the tendon usually has enough energy to sever the ulnar neurovascular bundle.

When the thumb is fully extended, the depression of the anatomical snuff-box is seen on the lateral aspect of the wrist immediately distal to the radial styloid process (see Fig. 46.30C). The snuff-box is bounded on its anterolateral (radial) side by the tendons of abductor pollicis longus (laterally) and extensor pollicis brevis (immediately medial), and on its posteromedial/ulnar side by the tendon of extensor pollicis longus. Running a finger along extensor pollicis brevis enables palpa-

tion of the superficial radial nerve, which can be rolled from side to side on the tendon. The cephalic vein passes over the roof of the snuff-box, where it is visible and palpable, and the pulsation of the radial artery can be palpated deeply on its floor.

The tendons located within the six extensor compartments of the wrist are subcutaneous and palpable (see Fig. 46.30B). Compartment 1 (containing abductor pollicis longus and extensor pollicis brevis) sits over the lateral radius. Regional tenderness to palpation occurs in de Quervain's tenosynovitis. Compartment 2 (containing extensors carpi radialis longus and brevis) and 3 (containing extensor pollicis longus) flank the lateral and medial sides of the dorsal radial (Lister's) tubercle, respectively. Compartments 5 (containing extensor digiti minimi) and 6 (containing extensor carpi ulnaris) flank the posterior and medial sides of the ulnar head, respectively. Compartment 4 (containing extensors digitorum and indicis) sits between the ulnar head and compartment 3. The tendons of extensors carpi radialis longus and brevis can be identified on the dorsal carpus when the fist is clenched and relaxed. The tendons of extensor digitorum are visible and palpable on the dorsal hand with the fingers fully extended. The tendon of extensor carpi ulnaris is palpable distal to the ulnar styloid as it crosses the wrist, with the wrist in extension and adduction. The first dorsal interosseous forms a visible fleshy mass on the lateral part of the dorsal hand between the index finger and thumb, especially whilst abducting the index finger against resistance.

The first metacarpal is both visible and palpable on the dorsolateral hand. Its dorsal aspect forms the lateral border of the hand. The fleshy elevation of the thenar eminence, produced by abductor and flexor pollicis brevis overlying opponens pollicis, is located on the lateral palm (see Fig. 46.30A). The less prominent hypothenar eminence is located on the medial palm and is formed by the comparable muscles of the little finger. The medial aspect of the hypothenar eminence forms the medial border of the hand.

VESSELS, PULSES AND NERVES

Arteries

The brachial artery travels along the medial arm in the depression posterior to coracobrachialis and along the furrow to the medial side of biceps, to a point just medial to the biceps tendon and level with the radial head/neck (see Fig. 46.29A–B). Its pulsations are palpable along this line of travel and proximal to the bicipital aponeurosis, *via* compression against the humerus. Brachial artery catheterization just proximal to the aponeurosis is used for coronary angiography or cardiac catheterization. The ability to compress the brachial artery against the humerus makes the arm the favoured site for non-invasive blood pressure measurement. The median nerve is intimately related to the brachial artery throughout its course in the arm. Proximally, it sits lateral to the artery; then, around the midpoint of the arm, it passes anterior to the artery and descends on its medial side to the cubital fossa.

The radial and ulnar arteries originate in the anterior forearm at a point situated medial to the biceps tendon and level with the lower head/neck of the radius (see Fig. 46.29B). From here, the ulnar artery passes inferomedially to a point one-third of the way down flexor carpi ulnaris (which passes along a line joining the medial humeral epicondyle to the pisiform). The ulnar artery and nerve then pass inferiorly to a point just lateral to the pisiform along the line of flexor carpi ulnaris, with the ulnar artery lying on the lateral (radial) side of the nerve. The radial artery passes inferolaterally through the forearm, from its point of origin to a point just lateral to the tendon of flexor carpi radialis, anterior to the distal radius. The upper part of its course can be represented by a line that passes deep to the medial part of the elevation produced by brachioradialis on the anterior aspect of the forearm. It then passes posteriorly, deep to the tendons of abductor pollicis longus and extensor pollicis brevis, and enters the anatomical snuff-box, where its pulsations are palpable.

The radial pulse is palpable on the anterior (volar) aspect of the wrist between the tendon of flexor carpi radialis medially and the radial styloid process laterally. This is the simplest pulse to palpate during routine patient examination. The pulse of the ulnar artery is usually palpable on deep pressure lateral to the tendon of flexor carpi ulnaris at the wrist.

The position of the superficial palmar arterial arch is indicated by a horizontal line, approximately 4 cm long, drawn at the level of the fully extended and partially abducted thumb (see Fig. 46.30A). A horizontal line, approximately 4 cm long, running from a point just distal to the hook of the hamate, indicates the position of the deep palmar arterial arch. The deep arch is located approximately 1 cm proximal to the superficial arch. The digital neurovascular bundles pass along the medial and lateral sides of the dorsal and volar surfaces of the digits, and can be anaesthetized in these positions *via* the web space.

Sites for arterial cannulation

At the wrist, the radial artery is commonly used for arterial cannulation for arterial blood sampling and blood pressure monitoring. Thrombosis or vasospasm of the radial artery at this site should not normally jeopardize the circulation to the hand since the interconnecting palmar arterial arches provide a collateral circulation to the hand. Allen's test should be performed prior to any such procedure.

Allen's test

Allen's test examines the patency of the interconnection between the superficial and deep palmar arches, and determines whether either artery is capable of maintaining the arterial supply to the hand in isolation. To perform the test, the radial and ulnar arteries of one hand are compressed by the examiner's thumbs. The forearm is then elevated and the hand exsanguinated by clenching the fist. The hand is then opened while arterial compression is continued. The palm of the hand appears pale, unless there is an anomalous arterial supply, e.g. from a persistent median artery. The rapid return of a normal pink colour to the palmar skin following removal of digital pressure from one of the arteries suggests that that artery is capable of perfusing the hand adequately in isolation. If the palmar skin remains pale, it must be assumed either that there is a distal obstruction to the artery or that the artery cannot adequately perfuse the hand in isolation. The test is then repeated with release of digital pressure on the other artery.

Veins

The superficial veins of the upper limb are usually visible and palpable, especially when a tourniquet occludes venous return proximally. At the

wrist, the cephalic vein consistently passes over the roof of the anatomical snuff-box (see Fig. 46.30C) and the dorsolateral aspect of the distal radius. Here, it sits next to the superficial branch of the radial nerve. The cephalic vein ascends the anterolateral forearm and the lateral arm in the groove lateral to biceps brachii. Superiorly, it ascends the deltopectoral groove to the infraclavicular fossa, where it pierces the clavipectoral fascia to enter the axillary vein.

The median cubital and basilic veins may be identified over the cubital fossa. In the proximal forearm, the cephalic vein runs alongside the lateral cutaneous nerve of the forearm. The veins are frequently covered by fat, especially in the female, which makes them difficult to see. The median cubital vein usually arises from the cephalic vein approximately 2.5 cm distal to the lateral epicondyle of the humerus, and runs upwards and medially over the bicipital aponeurosis to join the basilic vein approximately 2.5 cm above the transverse crease of the elbow. The median vein of the forearm drains the venous plexuses on the palmar surface of the hand. It ascends on the front of the forearm and usually ends in either the basilic vein or the median cubital vein; sometimes, it divides, just distal to the elbow, into two branches, one of which joins the basilic vein and the other the cephalic vein.

Venepuncture at the elbow

Commonly accessed superficial veins include the cephalic or median cubital vein at the cubital fossa, and the dorsal venous network of the hand. The cephalic vein may also be accessed over the distal radius and anatomical snuff-box. The latter site has many advantages for placing an indwelling cannula when a lengthy period is contemplated, although the vein lies close to the superficial branch of the radial nerve.

A high bifurcation of the brachial artery is seen occasionally and may produce an anomalous superficial ulnar artery that descends superficial to the common origin of the forearm flexor muscles. When this occurs, the ulnar artery, although deep to the deep fascia, may lie subcutaneously and pass superficial to the bicipital aponeurosis. Such an artery is therefore at risk of puncture during median cubital vein access.

Lymph nodes

Supratrochlear lymph nodes (in reality, supra-epicondylar nodes) lie next to the basilic vein. If enlarged, they may be palpated along the line of the vein, a few centimetres above the elbow joint.

Nerves

The trunks of the brachial plexus lie in the posterior triangle of the neck posterior to the subclavian artery (whose pulsations are palpable), in the angle between the clavicle and the lower posterior border of the clavicular head of sternocleidomastoid. The upper trunk can be palpated by passing the fingers inferiorly along the vertical groove between scalenus anterior and scalenus medius.

The axillary nerve and circumflex humeral vessels pass around the posterior and lateral aspects of the proximal humerus, close to the surgical neck, approximately 5 cm from the posterior edge, and 6 cm from the anterior edge, of the acromion (Cetik et al 2006). Bono et al (2000) found the axillary nerve to lie approximately 6 cm from the proximal humerus, passing within approximately 2 cm of the surgical neck of the humerus (see Fig. 46.29C).

The radial nerve passes inferolaterally along the spiral groove on the posterior arm (see Fig. 46.29C). The distance from the acromion to a line passing through the humeral epicondyles can be used as a baseline measurement; the radial nerve reaches the spiral groove approximately 46%, leaves it approximately 60%, and pierces the intermuscular septum approximately 66% of the way down this line (Cox et al 2010, Cho et al 2013). The radial nerve then passes inferiorly, approximately 1 cm lateral to the tendon of biceps, and bifurcates into the posterior interosseous and superficial radial nerves just distal to the lateral epicondyle (Cox et al 2010) (see Fig. 46.29B). The posterior interosseous nerve winds around the radial neck approximately 2.5 cm distal to the radial head (Cox et al 2010); this is an important surgical consideration when making an incision to expose the radial head. The nerve then passes inferiorly through the posterior forearm to a point midway between the ulnar head and the dorsal radial tubercle. The superficial radial nerve passes inferiorly, deep to brachioradialis, and becomes superficial passing between brachioradialis and extensor carpi radialis longus, approximately 8.5 cm proximal to the radial styloid process (Auerbach et al 1994, Park et al 2010). Its terminal branches are palpable as they pass over the tendon of extensor pollicis longus where it borders the anatomical snuff-box.

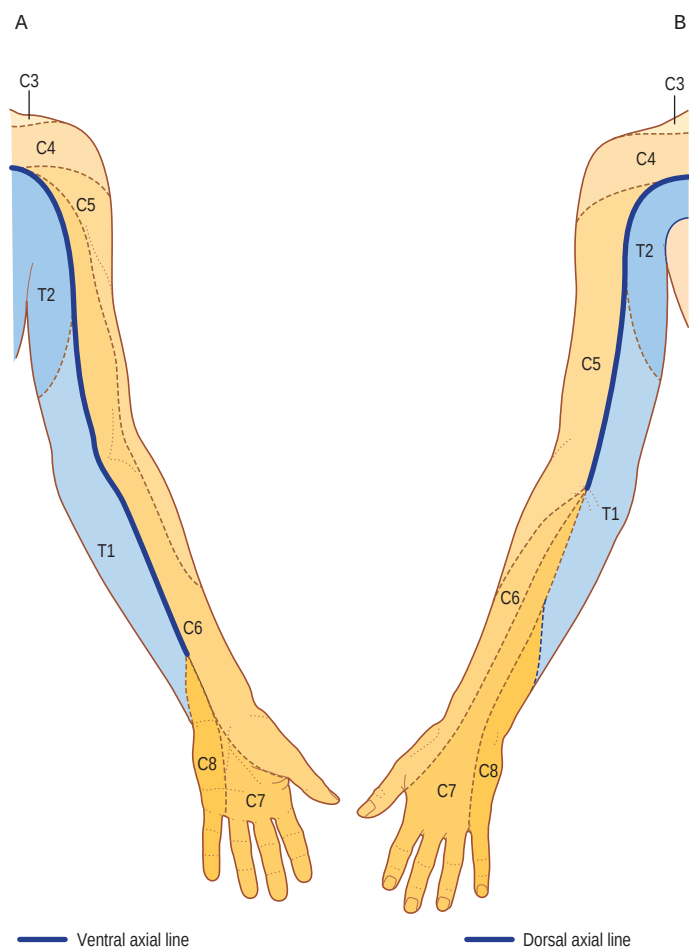


Fig. 46.31 The dermatomes of the left upper limb. Overlap across the axial lines is minimal, whereas overlap is considerable across the interrupted lines. **A**, Viewed from the anterior aspect. **B**, Viewed from the posterior aspect. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

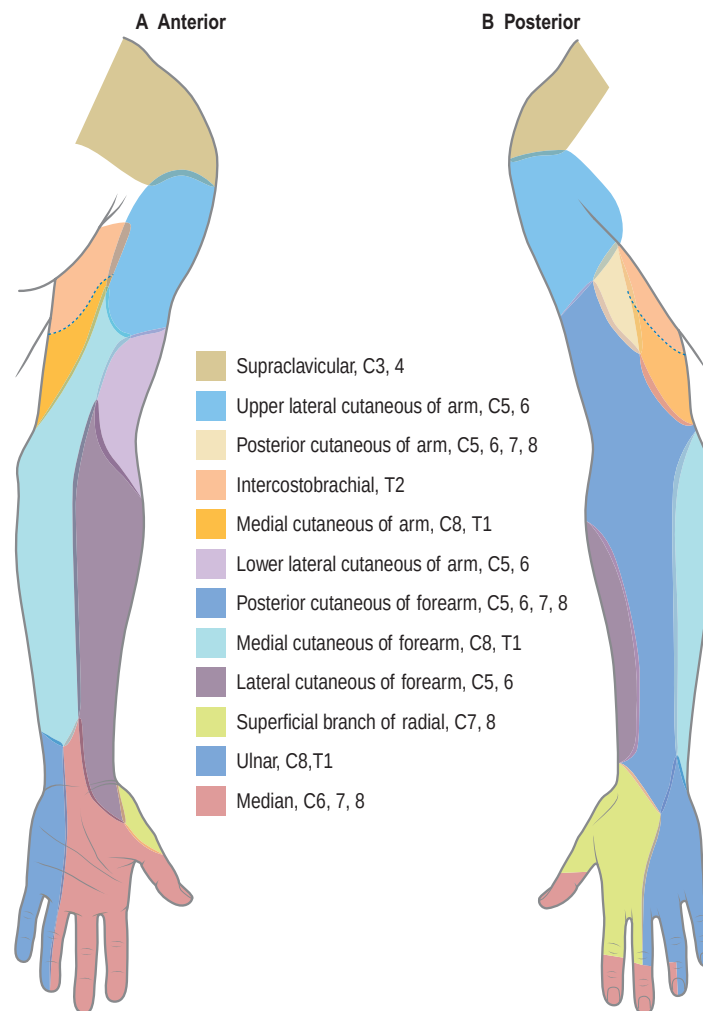


Fig. 46.32 The distribution of skin territories supplied by the cutaneous nerves of the left upper limb. **A**, Viewed from the anterior aspect. **B**, Viewed from the posterior aspect.

The median nerve is intimately related to the brachial artery throughout its course in the arm. In the forearm, it passes along a line joining a point just medial to the pulsating brachial artery in the cubital fossa to the tendon of palmaris longus, or to a point just lateral to the tendon at the wrist (see Fig. 46.29B).

The ulnar nerve passes posterior to the medial epicondyle and can be rolled from side to side just proximal to the epicondyle (see Fig. 46.29B). In the forearm and proximal hand, its course corresponds to a line drawn from the base of the posterior aspect of the medial epicondyle of the humerus to the lateral side of the pisiform, where it passes through Guyon's canal and across the hook of the hamate. Deep pressure at these bony landmarks can produce paraesthesia in the distribution of the nerve. Proximally, the nerve sits deep to flexor carpi ulnaris; distally, it sits lateral to the tendon of flexor carpi ulnaris and medial to the ulnar artery and its venae comitantes.

Dermatomes

Knowledge of the extent of individual dermatomes is largely based on clinical evidence. The dermatomes of the upper limb arise from spinal nerves C5–8 and T1; C7 supplies the central part of the hand (Figs 46.31–46.32). Considerable overlap exists between adjacent dermatomes innervated by nerves derived from consecutive spinal cord segments.

Myotomes

The predominant segmental origin of the nerve supply for each of the muscles of the upper limb and for the movements that take place at the joints of the upper limb are summarized in Tables 46.1–46.4.

Reflexes

Biceps jerk (C5, 6) The elbow is flexed to a right angle and slightly pronated. A finger is placed on the biceps tendon and struck with a percussion hammer; this should elicit flexion and slight supination of the forearm.

Triceps jerk (C6–8) The arm is supported at the wrist and flexed to a right angle. Triceps tendon is struck with a percussion hammer just proximal to the olecranon; this should elicit extension of the elbow.

Radial jerk (C7, 8) The radial jerk is a periosteal, not a tendon, reflex. The elbow is flexed to a right angle and the forearm placed in the mid pronation/supination position. The radial styloid is struck with the percussion hammer. This elicits contraction of brachioradialis, which causes flexion of the elbow.

Nerve blocks

Brachial plexus There are three common approaches to achieve anaesthetic blockade of the brachial plexus: the interscalene, supraclavicular and axillary routes.

The interscalene route requires identification of the interscalene groove. The patient is asked to sniff, an action that involves the scalene muscles as accessory muscles of respiration. A needle is inserted perpendicular to the skin and enters the interscalene groove. Paraesthesia may be elicited, and local anaesthetic solution is injected. There is a risk that some local anaesthetic may move in a retrograde direction and reach the cervical epidural space. The proximal nature of the block means that there is almost always some involvement of the phrenic nerve, and sympathetic block (as evidenced by a Horner's syndrome) is universal.

The supraclavicular route places local anaesthetic solution in the plane occupied by the trunks of the brachial plexus as they emerge between scalenus anterior and medius on the first rib, immediately posterior to the third part of the subclavian artery. The site of injection is 2 cm above the midpoint of the clavicle. The needle is directed backwards, inwards and downwards to make contact with the upper surface of the first rib. During this procedure, the patient will usually complain of paraesthesia down the arm, which indicates that the needle is correctly placed; preliminary aspiration ensures that a major vessel has not been punctured. If the first rib is missed, there is the risk of producing

a pneumothorax. There is often a transient Horner's syndrome, caused by diffusion of the local anaesthetic towards the stellate ganglion.

The axillary approach blocks the nerves as they group around the axillary artery.

Wrist Useful anaesthesia of the palm of the hand (except for the lateral surface of the thumb base) can be achieved by blockade of the ulnar and median nerves at the wrist (see **Video 50.2**). The ulnar nerve may be blocked via a needle inserted lateral to the tendon of flexor carpi ulnaris at the wrist. The median nerve may be blocked via a needle inserted at the midline between the tendons of palmaris longus and flexor carpi radialis at the level of the proximal wrist crease.

The radial nerve, which supplies the dorsal skin of the hand, may be blocked by injecting local anaesthetic around the subcutaneous

dorsoradial border of the wrist, where branches of the nerve can be rolled over the tendon of extensor pollicis longus (see **Fig. 46.30C**).

Digital nerves

The digital nerves lie on either side of the flexor sheath in a plane immediately anterior to the phalanx (see **Fig. 46.30A**). Therefore, the needle should be inserted on either side of the base of the digit, just anterior to the anterolateral margin of the phalanx. A common alternative approach to blocking the digital nerves is via a dorsal skin approach; the needle is inserted through the dorsal skin on either side of the base of the proximal phalanx and advanced towards the volar location of the digital nerves while the local anaesthetic agent is continually infused. This is thought to be less painful than the volar approach because the dorsal skin is less sensitive than the volar skin.

Bonus e-book images, tables and video

Fig. 46.17 The area of sensory abnormality after transection of C8 and T1.

Fig. 46.18 The area of sensory abnormality after section of the lateral cord in the axilla.

Fig. 46.19 Rupture of the posterior divisions of the trunks of the brachial plexus deep to the clavicle.

Fig. 46.20 Rupture of the axillary nerve.

Fig. 46.21 Sensory loss in two cases of rupture of the musculocutaneous nerve.

Fig. 46.22 A–B, Two examples of high lesion of the radial nerve.

Fig. 46.23 A–B, Transection of the median nerve, ulnar nerve, medial cutaneous nerve of the forearm, and brachial artery, in the arm of a 14-year-old boy.

Fig. 46.24 A high median nerve injury; there is no active flexion of the index finger and thumb.

Fig. 46.25 Section of the median nerve and the palmar cutaneous nerve at the wrist.

Fig. 46.26 A–B, A typical area of loss of sensibility after division of the nerve at the wrist, sparing the palmar cutaneous branch.

Fig. 46.27 The area of sensory loss and 'clawing' of little and ring fingers after division of the ulnar nerve in the forearm.

Fig. 46.28 Sympathetic paralysis seen within a few days of transection.

Table 46.1 Movements, muscles and segmental innervation in the upper limb.

Table 46.2 Segmental innervation of the muscles of the upper limb.

Table 46.3 Segmental innervation of joint movements of the upper limb.

Table 46.4 Movements and muscles tested to determine the location of a lesion in the upper limb.

Video 46.1 Upper limb surface anatomy. Clinical examination of muscles in the limbs.

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